Numerical PDEs

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Introduction

Partial differential equations arise naturally whenever a physical or mathematical quantity depends simultaneously on several independent variables. Heat diffusion, wave propagation, fluid flow, elasticity, electrostatics, and many other phenomena are described by PDEs. Although explicit solutions are available for a few special equations and geometries, most practical PDE problems cannot be solved analytically. Numerical methods therefore provide a systematic way to replace a continuous differential problem by a finite-dimensional algebraic problem that can be solved on a computer.

A central theme of numerical PDEs is the passage from the continuous problem to a discrete one. For example, a derivative may be approximated by nearby function values,

$$u'(x) \approx \frac{u(x+h) - u(x)}{h}, \qquad u''(x) \approx \frac{u(x-h) - 2u(x) + u(x+h)}{h^2}.$$

Such formulas are not merely computational devices. Their usefulness depends on how accurately they approximate the original differential operator and on how errors behave after the discrete equations are solved. This leads naturally to the basic concepts of *local truncation error*, *global error*, *consistency*, *stability*, and *convergence*. Throughout these notes, these ideas form the main framework for understanding and comparing numerical schemes.

The first part of the notes develops finite difference methods. Starting from Taylor expansions and polynomial interpolation, finite difference formulas for derivatives are constructed and their orders of accuracy are analyzed. These formulas are then applied to boundary value problems and elliptic equations. The resulting numerical schemes lead to linear systems

$$AU = F,$$

and the approximation error is often governed by an equation of the form

$$AE = \tau,$$

where E denotes the global discretization error and τ denotes the local truncation error. This simple relation already reveals one of the fundamental ideas of numerical analysis:

small truncation error alone is not sufficient;

the inverse discrete operator must also remain under control. In this sense, consistency describes how well the discrete equation approximates the PDE, while stability determines whether local errors are amplified. Their combination leads to convergence.

For elliptic problems, several complementary viewpoints appear. Eigenvalue analysis provides quantitative stability estimates in discrete L^2 norms, while Green's functions and discrete maximum principles lead naturally to L^∞ estimates. The maximum principle is particularly important because it connects the qualitative behavior of the continuous PDE with corresponding properties of its discrete approximation. Boundary conditions, variable coefficients, nonuniform meshes, and conservative discretizations further illustrate that the design of a numerical method must reflect the structure of the original differential operator.

A separate chapter is devoted to the *finite volume method*. This topic was **not covered in the lectures** on which the main body of these notes is based. It was studied independently after class and is included here as a supplementary part of the notes. The

finite volume method begins from the integral form of a conservation law rather than directly approximating derivatives pointwise. By integrating over computational cells and approximating fluxes across cell interfaces, one obtains schemes that preserve conservation at the discrete level. This viewpoint is especially natural for conservation laws and also provides a useful conceptual bridge between finite difference methods and numerical methods for hyperbolic PDEs.

The second major approach developed in these notes is based on weak formulations. Instead of requiring a PDE to hold pointwise, one tests the equation against suitable functions and transfers derivatives through integration by parts. This leads to weak derivatives, Sobolev spaces, and variational formulations. The Lax–Milgram lemma provides a general framework for existence and uniqueness, while the Galerkin method replaces an infinite-dimensional variational problem by a finite-dimensional one. The finite element method then emerges as a concrete realization of the Galerkin idea using piecewise polynomial basis functions. Céa's lemma shows that, under appropriate continuity and coercivity assumptions, the finite element solution is essentially as accurate as the best approximation available in the chosen discrete space.

Time-dependent problems introduce another source of discretization. Before studying parabolic and hyperbolic PDEs, it is therefore useful to review numerical methods for ordinary differential equations, including forward Euler, backward Euler, trapezoidal methods, and Runge–Kutta methods. These schemes illustrate again the distinction between consistency, stability, and convergence, but also introduce concepts such as absolute stability and stability regions.

For parabolic equations, spatial discretization and time discretization interact. Explicit schemes are often simple but conditionally stable, whereas implicit schemes may require the solution of a linear system at every time step but allow much larger time steps. Fourier analysis, energy estimates, maximum principles, operator splitting, and Galerkin semidiscretization provide several different tools for understanding these methods.

Hyperbolic equations behave differently. Information propagates along characteristic directions, and a stable numerical scheme must respect this propagation. This leads to upwind methods, Lax–Friedrichs and Lax–Wendroff schemes, numerical viscosity, and the CFL condition. For conservation laws, the conservative form of the numerical scheme becomes essential, since the discrete method should reproduce the same balance principle as the original PDE.

The purpose of these notes is therefore not merely to collect numerical formulas. The main goal is to understand why a discretization works, what controls its error, and which properties of the continuous PDE survive after discretization. Across finite difference, finite volume, finite element, parabolic, and hyperbolic methods, the recurring questions remain the same:

$$\text{How is the PDE discretized?} \qquad \text{How large is the error?}$$

$$\text{Is the scheme stable?} \qquad \text{Does the numerical solution converge?}$$

These questions provide the common structure underlying the numerical analysis of partial differential equations.

Chapter 1

Finite Difference Approximations and Error Analysis

1.1 Finite Difference Approximations and Order of Accuracy

Let $u(x)$ be a sufficiently smooth function. Our first goal is to approximate derivatives of u using values of u at a finite number of grid points.

Consider a uniform grid

$$x_j = x_0 + jh,$$

where $h > 0$ is the mesh size.

Definition 1.1.1 (Forward, Backward, and Central Differences)**.** The forward difference, backward difference, and central difference operators are defined by

$$D_+u(x) = \frac{u(x+h) - u(x)}{h},$$

$$D_-u(x) = \frac{u(x) - u(x-h)}{h},$$

and

$$D_0u(x) = \frac{u(x+h) - u(x-h)}{2h},$$

respectively.

We may also use wider centered differences. For example,

$$D_{0,2}u(x) = \frac{u(x+2h) - u(x-2h)}{4h}.$$

The main question is how accurately these discrete operators approximate the corresponding continuous derivative.

Suppose that $D_hu(x)$ is a numerical approximation of $u'(x)$. Define the error by

$$E(h) = \left|u'(x) - D_hu(x)\right|.$$

If

$$E(h) \approx Ch^p,$$

then p is called the *order of accuracy* of the approximation.

For the standard finite difference operators,

$$D_+,\ D_- \qquad \text{are first-order accurate,}$$

while

$$D_0,\ D_{0,2} \qquad \text{are second-order accurate.}$$

Taking logarithms gives

$$\log E(h) \approx \log C + p \log h.$$

Thus, in a log–log plot of $E(h)$ against h, the slope is approximately the order p.

In practice, $u'(x)$ may be unknown. We can still estimate the convergence order numerically.

Take $h_0 \ll h$ and regard $D_{h_0}u(x)$ as a highly accurate reference value. Define

$$E_1 = \left|D_hu(x) - D_{h_0}u(x)\right|.$$

If

$$D_h u(x) - u'(x) \approx Ch^p,$$

then

$$E_1 \approx C\left(h^p - h_0^p\right).$$

Similarly,

$$E_2 = \left|D_{h/2}u(x) - D_{h_0}u(x)\right| \approx C\left[\left(\frac{h}{2}\right)^p - h_0^p\right].$$

When $h_0 \ll h$,

$$E_1 \approx Ch^p, \qquad E_2 \approx C\left(\frac{h}{2}\right)^p.$$

Therefore,

$$\frac{E_1}{E_2} \approx 2^p,$$

and hence

$$\boxed{p \approx \log_2\left(\frac{E_1}{E_2}\right)}.$$

Remark 1.1.2. This provides a practical way to estimate the order of a numerical method even when the exact derivative is unavailable.

1.2 Taylor Expansion and Truncation Error

The order of a finite difference approximation can be derived systematically using Taylor expansion.

Assume first that u is sufficiently smooth. We have

$$u(x+h) = u(x) + hu'(x) + \frac{h^2}{2}u''(x) + \frac{h^3}{6}u^{(3)}(x) + O(h^4).$$

Therefore,

$$D_+u(x) = \frac{u(x+h) - u(x)}{h} = u'(x) + \frac{h}{2}u''(x) + O(h^2).$$

Hence,

$$D_+u(x) = u'(x) + O(h),$$

so the forward difference is first-order accurate.

Similarly,

$$u(x-h) = u(x) - hu'(x) + \frac{h^2}{2}u''(x) - \frac{h^3}{6}u^{(3)}(x) + O(h^4).$$

Therefore,

$$D_-u(x) = u'(x) - \frac{h}{2}u''(x) + O(h^2).$$

Thus the backward difference is also first-order accurate.

For the central difference,

$$D_0u(x) = \frac{u(x+h) - u(x-h)}{2h}.$$

Subtracting the two Taylor expansions gives

$$D_0u(x) = u'(x) + \frac{h^2}{6}u^{(3)}(x) + O(h^4).$$

Hence,

$$D_0u(x) = u'(x) + O(h^2),$$

so the central difference is second-order accurate.

More precisely, the mean-value form of the remainder gives

$$D_+u(x) = u'(x) + \frac{h}{2}u''(\xi_1)$$

for some

$$\xi_1 \in (x, x+h).$$

Similarly,

$$D_0u(x) = u'(x) + \frac{h^2}{12}\left[u^{(3)}(\xi_1) + u^{(3)}(\xi_2)\right],$$

where

$$\xi_1 \in (x, x+h), \qquad \xi_2 \in (x-h, x).$$

Remark 1.2.1 (Smoothness and Order). The formal order of a finite difference formula depends on the regularity of the exact solution. A high-order cancellation obtained from Taylor expansion is useful only when the required derivatives exist and remain bounded.

Thus, a formally second-order scheme may exhibit a lower observed order when the solution is not sufficiently smooth.

1.3 Polynomial Interpolation and General Difference Formulas

Finite difference formulas can also be derived from polynomial interpolation.

Let

$$x_0, x_1, \ldots, x_n$$

be distinct interpolation points, and suppose that the corresponding function values

$$f(x_0), f(x_1), \ldots, f(x_n)$$

are known.

The Lagrange interpolation polynomial is

$$P(x) = \sum_{i=0}^{n} f(x_i)L_i(x),$$

where

$$L_i(x) = \prod_{\substack{0\le j\le n\\ j\neq i}} \frac{x - x_j}{x_i - x_j}.$$

The basis polynomials satisfy

$$L_i(x_j) = \delta_{ij}.$$

Theorem 1.3.1 (Interpolation Error)**.** *Suppose*

$$f \in C^{n+1}.$$

Then, for every x in the interval containing the interpolation points, there exists some $\xi(x)$ such that

$$f(x) = P(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} \prod_{j=0}^{n} (x - x_j).$$

To approximate $f'(x_k)$, differentiate the interpolation polynomial and define

$$Df(x_k) := P'(x_k).$$

Since

$$P'(x_k) = \sum_{i=0}^{n} f(x_i) L_i'(x_k),$$

we obtain a finite difference formula of the form

$$Df(x_k) = \sum_{i=0}^{n} c_i f(x_i),$$

where

$$c_i = L_i'(x_k).$$

Differentiating the interpolation error gives

$$f'(x_k) = P'(x_k) + \frac{f^{(n+1)}(\xi_k)}{(n+1)!} \prod_{\substack{0 \le j \le n \\ j \ne k}} (x_k - x_j),$$

for a suitable point ξ_k. Therefore,

$$Df(x_k) = f'(x_k) - \frac{f^{(n+1)}(\xi_k)}{(n+1)!} \prod_{\substack{0 \le j \le n \\ j \ne k}} (x_k - x_j).$$

This provides another systematic derivation of finite difference formulas and their truncation errors.

Difference operators can also be combined to approximate higher-order derivatives. For example,

$$D_+ D_- u(x) = \frac{u(x-h) - 2u(x) + u(x+h)}{h^2}.$$

This is the standard centered approximation of the second derivative:

$$D_+ D_- u(x) = u''(x) + O(h^2).$$

In operator notation, if the shift variable is represented formally by x, the numerators of several difference operators have the following forms:

$$\begin{aligned} D_+ : &\qquad x - 1, \\ D_- : &\qquad 1 - x^{-1}, \\ D_2 = D_+ D_- : &\qquad x - 2 + x^{-1}, \\ D_0 : &\qquad x - x^{-1}, \\ D_{+,2} : &\qquad x^2 - 4x + 3, \end{aligned}$$

and

$$D_3 = D_+ D_- D_0 : \qquad x^2 - 2x + 2x^{-1} - x^{-2}.$$

Remark 1.3.2. These symbolic expressions are useful for understanding how finite difference operators are composed. The corresponding powers of the shift operator encode which neighboring grid points appear in the finite difference stencil.

We now consider the general problem of constructing a finite difference approximation. Let

$$x_0, x_1, \ldots, x_n$$

be grid points near a reference point $\bar{x}$. We seek coefficients

$$c_0, c_1, \ldots, c_n$$

such that

$$D^k u(\bar{x}) := \sum_{j=0}^{n} c_j u(x_j) = u^{(k)}(\bar{x}) + O(h^p),$$

where the order p is as large as possible.

Expand each $u(x_j)$ about $\bar{x}$:

$$u(x_j) = u(\bar{x}) + (x_j - \bar{x})u'(\bar{x}) + \cdots$$

$$+\frac{(x_j - \bar{x})^k}{k!}u^{(k)}(\bar{x}) + \cdots .$$

Substituting these expansions gives

$$\sum_{j=0}^{n} c_j u(x_j) = \sum_{i=0}^{\infty} \left[\frac{1}{i!} \sum_{j=0}^{n} c_j (x_j - \bar{x})^i \right] u^{(i)}(\bar{x}).$$

To reproduce $u^{(k)}(\bar{x})$, the coefficients should satisfy the moment conditions

$$\frac{1}{i!} \sum_{j=0}^{n} c_j (x_j - \bar{x})^i = \begin{cases} 1, & i = k, \\ 0, & i \neq k, \end{cases}$$

for as many values of i as possible.

Using $n + 1$ grid points, we generally expect

$$p \geq n + 1 - k.$$

Remark 1.3.3 (Symmetry and Superconvergence). The estimate

$$p \geq n + 1 - k$$

is a generic expectation. For symmetric stencils, additional Taylor terms may cancel automatically, producing an order higher than this basic count suggests.

The centered approximation

$$\frac{u(x - h) - 2u(x) + u(x + h)}{h^2}$$

is an important example: three points approximate a second derivative, yet symmetry produces second-order rather than merely first-order accuracy.

1.4 Local Truncation Error and Global Error

The previous sections concern the approximation of individual derivatives. When finite differences are used to discretize a differential equation, two different types of error must be distinguished.

Suppose a continuous problem is written abstractly as

$$Lu = f,$$

and its finite difference approximation is

$$L_h U_h = F_h.$$

Let u_h denote the exact solution sampled at the grid points.

Definition 1.4.1 (Local Truncation Error)**.** The *local truncation error* is the residual obtained by substituting the exact solution into the discrete equation:

$$\tau_h = L_h u_h - F_h.$$

Thus, the local truncation error measures how accurately the continuous differential equation is represented by the discrete equation.

Definition 1.4.2 (Global Error)**.** The *global error* is

$$E_h = u_h - U_h,$$

where U_h is the numerical solution.

For a linear problem,

$$L_h U_h = F_h$$

and

$$L_h u_h = F_h + \tau_h.$$

Subtracting gives the fundamental error equation

$$\boxed{L_h E_h = \tau_h}.$$

If L_h is invertible, then

$$E_h = L_h^{-1} \tau_h.$$

This identity explains why truncation error alone does not determine the global error.

1.5 Consistency, Stability, and Convergence

Definition 1.5.1 (Consistency)**.** A numerical discretization is *consistent* if its local truncation error tends to zero as the mesh is refined:

$$\|\tau_h\| \longrightarrow 0 \qquad \text{as } h \longrightarrow 0.$$

More precisely, if

$$\|\tau_h\| \leq C h^p,$$

where C is independent of h, then the discretization is said to be consistent of order p.

Consistency concerns the approximation of the *equation*, not directly the approximation of its solution.

For the discrete linear problem

$$L_h U_h = F_h,$$

stability controls the sensitivity of the numerical solution to perturbations in the discrete data.

Definition 1.5.2 (Stability)**.** The discretization is *stable* if there exist constants $h_0 > 0$ and $C > 0$, independent of h, such that

$$\|L_h^{-1}\| \leq C, \qquad 0 < h \leq h_0.$$

Equivalently, for

$$L_h U_h = F_h,$$

we have the uniform estimate

$$\|U_h\| \leq C\|F_h\|.$$

The important point is that C must be independent of the mesh size h.

Definition 1.5.3 (Convergence)**.** A numerical method is *convergent* if

$$\|E_h\| = \|u_h - U_h\| \longrightarrow 0 \qquad \text{as } h \longrightarrow 0.$$

Convergence concerns the approximation of the *solution.*

The distinction among the three basic concepts is therefore

Consistency :	Does the discrete equation approximate the PDE?
Stability :	Does the discrete problem control perturbations?
Convergence :	Does the numerical solution approach the exact solution?

The fundamental relation among these concepts can already be seen from the error equation

$$L_h E_h = \tau_h.$$

Theorem 1.5.4 (Consistency and Stability Imply Convergence)**.** *Suppose that the discrete problem is linear and that*

$$\|L_h^{-1}\| \leq C$$

uniformly for sufficiently small h. If the discretization is consistent, then it is convergent.

Proof. Since

$$L_h E_h = \tau_h,$$

we have

$$E_h = L_h^{-1}\tau_h.$$

Therefore,

$$\|E_h\| \leq \|L_h^{-1}\|\,\|\tau_h\| \leq C\|\tau_h\|.$$

By consistency,

$$\|\tau_h\| \longrightarrow 0 \qquad \text{as } h \longrightarrow 0.$$

Hence,

$$\|E_h\| \longrightarrow 0,$$

which proves convergence. □

Moreover, if

$$\|\tau_h\| \le C_1 h^p$$

and

$$\|L_h^{-1}\| \le C_2,$$

then

$$\|E_h\| \le C_1 C_2 h^p.$$

Thus, under uniform stability, an order-p truncation error gives an order-p global error.

Remark 1.5.5. The relation

$$\text{Consistency} + \text{Stability} \Longrightarrow \text{Convergence}$$

will repeatedly appear throughout these notes. The precise meaning of stability depends on the type of PDE and on the norm being considered, but the underlying principle remains the same.

1.6 Error Analysis Framework

The basic procedure for analyzing a finite difference method can be summarized as follows.

1. Construct a discrete approximation L_h of the differential operator L.
2. Substitute the exact solution into the discrete equation and determine the local truncation error
$$\tau_h = L_h u_h - F_h.$$
3. Determine the order of consistency, for example,
$$\|\tau_h\| = O(h^p).$$
4. Establish a stability estimate independent of h:
$$\|L_h^{-1}\| \le C.$$
5. Use the error equation
$$L_h E_h = \tau_h$$
to obtain
$$\|E_h\| \le C\|\tau_h\|.$$
6. Conclude convergence and determine the order of the global error.

This framework will be applied first to two-point boundary value problems and later to elliptic, parabolic, and hyperbolic partial differential equations.

Chapter 2

Finite Difference Methods for Boundary Value Problems

2.1 Two-Point Boundary Value Problems and Discretization

Consider the two-point boundary value problem

$$\begin{cases} u''(x) = f(x), & x \in (0,1), \\ u(0) = u(1) = 0. \end{cases}$$

Introduce the uniform grid

$$x_j = jh, \qquad j = 0, 1, \ldots, N, \qquad h = \frac{1}{N}.$$

Let

$$u_j \approx u(x_j).$$

Since the boundary values are known,

$$u_0 = u_N = 0.$$

The unknowns are

$$u_1, u_2, \ldots, u_{N-1}.$$

Using the centered difference approximation,

$$u''(x_j) \approx \frac{u_{j-1} - 2u_j + u_{j+1}}{h^2},$$

we obtain

$$\frac{u_{j-1} - 2u_j + u_{j+1}}{h^2} = f_j, \qquad f_j = f(x_j).$$

Therefore, the discrete problem can be written as

$$AU = F,$$

where

$$U = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{bmatrix}, \qquad F = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{N-1} \end{bmatrix},$$

and

$$A = \frac{1}{h^2} \begin{bmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -2 \end{bmatrix}.$$

Thus the continuous boundary value problem has been reduced to a finite-dimensional linear system.

Define the pointwise error by

$$e_j = u(x_j) - u_j,$$

and let

$$E = \begin{bmatrix} e_1 & e_2 & \cdots & e_{N-1} \end{bmatrix}^T.$$

For a grid function E, define the discrete p-norm by

$$\|E\|_p = \left(h \sum_{j=1}^{N-1} |e_j|^p \right)^{1/p}.$$

In particular,

$$\|E\|_1 = h \sum_{j=1}^{N-1} |e_j|,$$

$$\|E\|_2 = \left(h \sum_{j=1}^{N-1} |e_j|^2 \right)^{1/2},$$

and

$$\|E\|_\infty = \max_{1 \le j \le N-1} |e_j|.$$

For the exact solution, Taylor expansion gives

$$\frac{u(x_{j-1}) - 2u(x_j) + u(x_{j+1})}{h^2} = f(x_j) + \tau_j,$$

where

$$\tau_j = \frac{h^2}{12} u^{(4)}(\xi_j)$$

for some

$$\xi_j \in (x_{j-1}, x_{j+1}).$$

Thus,

$$\frac{e_{j-1} - 2e_j + e_{j+1}}{h^2} = \tau_j.$$

In matrix form,

$$AE = \tau.$$

Hence,

$$E = A^{-1}\tau.$$

2.2 L^2 Stability and Error Estimate

For a matrix M, the induced 2-norm satisfies

$$\|M\|_2 = \sqrt{\rho(M^T M)},$$

where ρ denotes the spectral radius. If M is symmetric, then

$$\|M\|_2 = \rho(M) = \max_{\lambda \in \sigma(M)} |\lambda|.$$

Since

$$A = A^T,$$

we have

$$\|A^{-1}\|_2 = \frac{1}{\min_{\lambda \in \sigma(A)} |\lambda|}.$$

Therefore, stability can be established by determining the eigenvalues of A.

Theorem 2.2.1 (Gershgorin Circle Theorem)**.** *Let $M = (m_{ij}) \in \mathbb{C}^{n\times n}$. Every eigenvalue of M lies in the union*

$$\bigcup_{i=1}^{n} R_i,$$

where

$$R_i = \left\{ z \in \mathbb{C} : |z - m_{ii}| \leq \sum_{j\neq i} |m_{ij}| \right\}.$$

For the finite difference matrix A, Gershgorin's theorem gives

$$-\frac{4}{h^2} \leq \lambda \leq 0.$$

To obtain the exact eigenvalues, suppose

$$Av = \lambda v.$$

Then

$$\frac{v_{j-1} - 2v_j + v_{j+1}}{h^2} = \lambda v_j.$$

Let

$$\alpha = h^2\lambda.$$

Then

$$v_{j-1} - (2 + \alpha)v_j + v_{j+1} = 0.$$

Since

$$-4 \leq \alpha \leq 0,$$

write

$$2 + \alpha = 2\cos\theta.$$

The characteristic roots are

$$\mu_{1,2} = e^{\pm i\theta}.$$

Therefore,

$$v_j = C_1 \cos(j\theta) + C_2 \sin(j\theta).$$

The boundary condition

$$v_0 = 0$$

gives

$$C_1 = 0.$$

The second boundary condition

$$v_N = 0$$

gives

$$\sin(N\theta) = 0.$$

Hence,

$$\theta_k = \frac{k\pi}{N}, \qquad k = 1, \ldots, N-1.$$

Therefore,

$$\lambda_k = \frac{2\cos(k\pi/N) - 2}{h^2},$$

or equivalently,

$$\lambda_k = -\frac{4}{h^2}\sin^2\left(\frac{k\pi}{2N}\right).$$

The corresponding eigenvectors are

$$v^{(k)} = \begin{bmatrix} \sin(k\pi/N) \\ \sin(2k\pi/N) \\ \vdots \\ \sin((N-1)k\pi/N) \end{bmatrix}.$$

Since

$$h = \frac{1}{N},$$

the smallest absolute eigenvalue is

$$\min_k |\lambda_k| = \frac{4}{h^2}\sin^2\left(\frac{\pi h}{2}\right).$$

Using

$$\sin x \geq \frac{2x}{\pi}, \qquad 0 \leq x \leq \frac{\pi}{2},$$

we obtain

$$\min_k |\lambda_k| \geq 4.$$

Consequently,

$$\boxed{\|A^{-1}\|_2 \leq \frac{1}{4}}.$$

Thus the finite difference scheme is stable in the discrete L^2-norm.

Suppose

$$u \in C^4[0,1].$$

Then

$$|\tau_j| \leq Ch^2,$$

where

$$C = \frac{1}{12}\max_{x\in[0,1]} |u^{(4)}(x)|.$$

Therefore,

$$\|\tau\|_2 \leq Ch^2.$$

Since

$$AE = \tau,$$

we have

$$\|E\|_2 \leq \|A^{-1}\|_2\|\tau\|_2.$$

Hence,

$$\boxed{\|E\|_2 \leq \frac{1}{4}Ch^2}.$$

Thus the centered finite difference method is second-order convergent in the discrete L^2-norm.

2.3 Green's Function and L^∞ Stability

The exact solution of

$$\begin{cases} u''(x) = f(x), & x \in (0,1), \\ u(0) = u(1) = 0 \end{cases}$$

can be represented using a Green's function.

For fixed $y \in (0,1)$, define $G(x,y)$ by

$$G_{xx}(x,y) = \delta(x-y),$$

with

$$G(0,y) = G(1,y) = 0.$$

Using the defining property of the Dirac delta function,

$$u(y) = \int_0^1 \delta(x-y)u(x)\,dx.$$

Since

$$\delta(x-y) = G_{xx}(x,y),$$

we obtain

$$u(y) = \int_0^1 G_{xx}(x,y)u(x)\,dx.$$

Integrating by parts twice and using the boundary conditions gives

$$u(y) = \int_0^1 G(x,y)u''(x)\,dx.$$

Therefore,

$$u(y) = \int_0^1 G(x,y)f(x)\,dx.$$

Define the Heaviside function by

$$H(x) = \int_{-\infty}^{x} \delta(s)\,ds.$$

Since

$$G_{xx}(x,y) = \delta(x-y),$$

integration gives

$$G_x(x,y) = H(x-y) + C_1.$$

Integrating once more and applying the boundary conditions and continuity at $x = y$, we obtain

$$\boxed{G(x,y) = \begin{cases} (y-1)x, & 0 \le x \le y, \\ y(x-1), & y \le x \le 1. \end{cases}}$$

Since

$$G(x,y) \le 0$$

for

$$x, y \in [0,1],$$

we have

$$f \geq 0 \quad \Longrightarrow \quad u \leq 0,$$

and

$$f \leq 0 \quad \Longrightarrow \quad u \geq 0.$$

Furthermore,

$$|u(y)| \leq \int_0^1 |G(x,y)|\,|f(x)|\,dx.$$

A direct calculation gives

$$\int_0^1 |G(x,y)|\,dx = \frac{1}{2}y(1-y) \leq \frac{1}{8}.$$

Therefore,

$$\boxed{\|u\|_\infty \leq \frac{1}{8}\|f\|_\infty}.$$

Define the solution operator

$$S[f](x) = \int_0^1 f(y)G(y,x)\,dy.$$

Then

$$u(x) = S[f](x).$$

At a grid point x_i, applying the composite trapezoidal rule gives

$$u(x_i) = h\sum_{j=1}^{N-1} G(x_j, x_i)f_j + O(h^2).$$

Define

$$B_{ij} = hG(x_i, x_j).$$

Then

$$\widetilde{U} = BF.$$

A direct calculation shows that

$$AB = I.$$

Hence,

$$\boxed{B = A^{-1}}.$$

Thus the inverse finite difference matrix can be interpreted as a discrete Green's function.

Since

$$B = A^{-1},$$

we have

$$\|A^{-1}\|_\infty = \max_i \sum_{j=1}^{N-1} |B_{ij}|.$$

Using the Green's function representation,

$$\sum_{j=1}^{N-1} |B_{ij}| \leq \int_0^1 |G(x,x_i)|\,dx.$$

Therefore,

$$\boxed{\|A^{-1}\|_\infty \le \frac{1}{8}}.$$

Since

$$E = A^{-1}\tau,$$

we obtain

$$\|E\|_\infty \le \|A^{-1}\|_\infty \|\tau\|_\infty.$$

If

$$u \in C^4[0,1],$$

then

$$\|\tau\|_\infty \le Ch^2.$$

Thus,

$$\boxed{\|E\|_\infty \le \frac{1}{8}Ch^2}.$$

Therefore, the scheme is second-order convergent in the discrete L^∞-norm.

2.4 Dirichlet and Neumann Boundary Conditions

Consider first the nonhomogeneous Dirichlet problem

$$\begin{cases} u''(x) = f(x), \quad x \in (a,b), \\ u(a) = \alpha, \\ u(b) = \beta. \end{cases}$$

The interior finite difference formula remains unchanged:

$$\frac{u_{j-1} - 2u_j + u_{j+1}}{h^2} = f_j.$$

The boundary values are moved to the right-hand side. For example,

$$\frac{\alpha - 2u_1 + u_2}{h^2} = f_1,$$

so

$$\frac{-2u_1 + u_2}{h^2} = f_1 - \frac{\alpha}{h^2}.$$

Similarly,

$$\frac{u_{N-2} - 2u_{N-1}}{h^2} = f_{N-1} - \frac{\beta}{h^2}.$$

Now consider a derivative boundary condition

$$u'(a) = \sigma.$$

A first-order approximation is

$$\frac{u_1 - u_0}{h} = \sigma.$$

A one-sided second-order approximation is

$$\frac{-3u_0 + 4u_1 - u_2}{2h} = \sigma.$$

Another approach is the ghost-point method. Introduce

$$x_{-1} = x_0 - h.$$

Use

$$\frac{u_1 - u_{-1}}{2h} = \sigma,$$

together with

$$\frac{u_{-1} - 2u_0 + u_1}{h^2} = f_0.$$

From the first equation,

$$u_{-1} = u_1 - 2h\sigma.$$

Substituting into the second gives

$$\frac{2u_1 - 2u_0 - 2h\sigma}{h^2} = f_0.$$

Hence,

$$u_1 - u_0 = h\sigma + \frac{h^2}{2} f_0.$$

Remark 2.4.1. The ghost-point method preserves a centered treatment of the derivative boundary condition. After elimination of the ghost value, the resulting matrix need not remain symmetric.

We now consider the pure Neumann problem

$$\begin{cases} u''(x) = f(x), \quad x \in (0,1), \\ u'(0) = \sigma, \\ u'(1) = \gamma. \end{cases}$$

After eliminating the ghost points, the resulting matrix is singular. After a convenient symmetrization, it has the form

$$A = \frac{1}{h^2} \begin{bmatrix} -1 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -1 \end{bmatrix}.$$

Indeed,

$$A\mathbf{1} = 0,$$

where

$$\mathbf{1} = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^T.$$

This agrees with the continuous problem: if $u(x)$ is a solution, then

$$u(x) + C$$

is also a solution.

The continuous compatibility condition follows from

$$\int_0^1 u''(x)\,dx = \int_0^1 f(x)\,dx.$$

Therefore,

$$\boxed{\int_0^1 f(x)\,dx = \gamma - \sigma}.$$

Since A is symmetric,

$$\mathcal{C}(A) = \mathcal{N}(A)^{\perp}.$$

Thus,

$$F \in \mathcal{C}(A) \quad \Longleftrightarrow \quad F^T\mathbf{1} = 0.$$

For the discretization in the notes, this becomes

$$\frac{h}{2}\left(f_0 + 2f_1 + \cdots + 2f_{N-1} + f_N\right) = \gamma - \sigma.$$

This is exactly the composite trapezoidal approximation of the continuous compatibility condition.

If U_s is one particular discrete solution, then every solution has the form

$$U = U_s + \alpha\mathbf{1}, \qquad \alpha \in \mathbb{R}.$$

A normalization condition can be imposed to determine a unique solution. For example, one may prescribe

$$\int_0^1 u(x)\,dx = C.$$

Its discrete trapezoidal analogue is

$$\frac{h}{2}\left(u_0 + 2u_1 + \cdots + 2u_{N-1} + u_N\right) = C.$$

This determines the constant α.

2.5 General and Conservative Discretizations

Consider the general second-order boundary value problem

$$\begin{cases} a(x)u''(x) + b(x)u'(x) + c(x)u(x) = f(x), & x \in (0,1), \\ u(0) = \alpha, \qquad u(1) = \beta. \end{cases}$$

Using centered finite differences,

$$u''(x_j) \approx \frac{u_{j-1} - 2u_j + u_{j+1}}{h^2},$$

and

$$u'(x_j) \approx \frac{u_{j+1} - u_{j-1}}{2h}.$$

Hence,

$$a(x_j)\frac{u_{j-1} - 2u_j + u_{j+1}}{h^2} + b(x_j)\frac{u_{j+1} - u_{j-1}}{2h} + c(x_j)u_j = f(x_j).$$

Provided the solution and coefficients are sufficiently smooth, this is a second-order discretization.

A particularly important problem is

$$\begin{cases} \Big(k(x)u'(x)\Big)' = f(x), & x \in (0,1), \\ u(0) = \alpha, \qquad u(1) = \beta. \end{cases}$$

Define the flux

$$q(x) = k(x)u'(x).$$

Approximate

$$q_{j+\frac{1}{2}} \approx k_{j+\frac{1}{2}} \frac{u_{j+1} - u_j}{h},$$

where

$$k_{j+\frac{1}{2}} = k\left(x_j + \frac{h}{2}\right).$$

Then

$$\Big(k(x)u'(x)\Big)'_{x=x_j}$$

is approximated by

$$\frac{q_{j+\frac{1}{2}} - q_{j-\frac{1}{2}}}{h}.$$

Therefore,

$$\frac{1}{h^2}\left[k_{j+\frac{1}{2}}(u_{j+1} - u_j) - k_{j-\frac{1}{2}}(u_j - u_{j-1})\right] = f_j.$$

Equivalently,

$$\frac{1}{h^2}\left[k_{j-\frac{1}{2}}u_{j-1} - \left(k_{j-\frac{1}{2}} + k_{j+\frac{1}{2}}\right)u_j + k_{j+\frac{1}{2}}u_{j+1}\right] = f_j.$$

The resulting matrix is symmetric. If the sign convention is changed to

$$-\Big(k(x)u'(x)\Big)',$$

and

$$k(x) > 0,$$

the corresponding matrix is positive definite.

Remark 2.5.1. The conservative form preserves the flux-balance structure of the continuous differential equation. This idea will reappear in the finite volume method.

2.6 Nonuniform Meshes

A uniform mesh is not always the most efficient choice. Consider

$$\begin{cases} \varepsilon u''(x) - u'(x) = f(x), & x \in (0,1), \\ u(0) = \alpha, \qquad u(1) = \beta. \end{cases}$$

For

$$f(x) \equiv -1,$$

the exact solution has the form

$$u(x) = \alpha + x + (\beta - \alpha - 1)\frac{e^{x/\varepsilon} - 1}{e^{1/\varepsilon} - 1}.$$

When

$$\varepsilon \ll 1,$$

the solution changes rapidly near

$$x = 1.$$

Indeed, derivatives of the solution become very large in this region. A uniform mesh may therefore require an unnecessarily large number of points throughout the whole interval.

A nonuniform mesh places more grid points in regions where the solution varies rapidly and fewer points where the solution is smooth. Let

$$x_0 < x_1 < \cdots < x_N,$$

and define

$$h_j = x_j - x_{j-1}.$$

A standard approximation to the second derivative is

$$u''(x_j) \approx \frac{2}{h_j + h_{j+1}} \left(\frac{u_{j+1} - u_j}{h_{j+1}} - \frac{u_j - u_{j-1}}{h_j} \right).$$

Thus nonuniform meshes allow the discretization to adapt to the spatial structure of the solution.

Remark 2.6.1. The example above contains a boundary layer near $x = 1$. When large derivatives are localized in a small region, mesh refinement should also be localized rather than distributed uniformly over the entire interval.

Chapter 3

Finite Difference Methods for Elliptic PDEs

3.1 Elliptic Equations and the Five-Point Scheme

Consider the general second-order equation

$$a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + b_1u_x + b_2u_y + cu = f(x, y).$$

The second-order part is determined by

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}.$$

The equation is elliptic when the two eigenvalues of A have the same sign. In two dimensions, this is equivalent to

$$a_{12}^2 - a_{11}a_{22} < 0.$$

A basic model problem is the Poisson equation

$$\begin{cases} \Delta u = f, & (x, y) \in \Omega, \\ u = 0, & (x, y) \in \partial\Omega, \end{cases}$$

where

$$\Delta u = u_{xx} + u_{yy}.$$

We first take

$$\Omega = [0, 1]^2.$$

Introduce the uniform grid

$$h = \frac{1}{N+1}, \qquad x_i = ih, \qquad y_j = jh.$$

Let

$$u_{ij} \approx u(x_i, y_j).$$

At an interior grid point,

$$u_{xx}(x_i, y_j) \approx \frac{u_{i-1,j} - 2u_{ij} + u_{i+1,j}}{h^2},$$

and

$$u_{yy}(x_i, y_j) \approx \frac{u_{i,j-1} - 2u_{ij} + u_{i,j+1}}{h^2}.$$

Therefore,

$$\Delta u(x_i, y_j)$$

is approximated by

$$\Delta_h u_{ij} = \frac{u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{ij}}{h^2}.$$

The finite difference scheme is

$$\boxed{\Delta_h u_{ij} = f_{ij}}, \qquad f_{ij} = f(x_i, y_j).$$

The corresponding five-point stencil is

$$\frac{1}{h^2} \begin{array}{ccc} & 1 & \\ 1 & -4 & 1 \\ & 1 & \end{array}.$$

3.2 Matrix Structure and Local Truncation Error

There are N^2 interior unknowns. Order them lexicographically as

$$(1,1),(1,2),\ldots,(1,N),$$

$$(2,1),(2,2),\ldots,(2,N),$$

and so on. Then the discrete problem has the form

$$AU = F.$$

Define

$$T = \begin{bmatrix} -4 & 1 & & \\ 1 & -4 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -4 \end{bmatrix}.$$

Then

$$A = \frac{1}{h^2}\begin{bmatrix} T & I & & \\ I & T & I & \\ & \ddots & \ddots & \ddots \\ & & I & T \end{bmatrix}.$$

The matrix has size

$$N^2 \times N^2,$$

but the number of nonzero entries is only approximately

$$5N^2.$$

Thus A is sparse.

Let

$$D = \frac{1}{h^2}\begin{bmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -2 \end{bmatrix}.$$

Then the two-dimensional discrete Laplacian can be written as

$$\boxed{A = I_N \otimes D + D \otimes I_N}.$$

This reflects the continuous decomposition

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

Let

$$\widehat{u}_{ij} = u(x_i, y_j)$$

denote the exact solution evaluated at the grid points. Taylor expansion gives

$$\frac{\widehat{u}_{i-1,j} - 2\widehat{u}_{ij} + \widehat{u}_{i+1,j}}{h^2} = u_{xx}(x_i, y_j) + \frac{h^2}{12}u_{xxxx}(x_i, y_j) + O(h^4),$$

and

$$\frac{\widehat{u}_{i,j-1} - 2\widehat{u}_{ij} + \widehat{u}_{i,j+1}}{h^2} = u_{yy}(x_i, y_j) + \frac{h^2}{12} u_{yyyy}(x_i, y_j) + O(h^4).$$

Therefore,

$$\Delta_h \widehat{u}_{ij} = \Delta u(x_i, y_j) + \tau_{ij},$$

where

$$\tau_{ij} = \frac{h^2}{12} \left(u_{xxxx}(x_i, y_j) + u_{yyyy}(x_i, y_j) \right) + O(h^4).$$

Hence the five-point scheme is second-order consistent.

Define

$$e_{ij} = \widehat{u}_{ij} - u_{ij}.$$

Since

$$\Delta_h u_{ij} = f_{ij}$$

and

$$\Delta_h \widehat{u}_{ij} = f_{ij} + \tau_{ij},$$

we obtain

$$\Delta_h e_{ij} = \tau_{ij}.$$

In matrix form,

$$\boxed{AE = \tau}.$$

3.3 L^2 Stability and Error Estimate

Since

$$A = A^T,$$

we have

$$\|A^{-1}\|_2 = \frac{1}{\min_{\lambda \in \sigma(A)} |\lambda|}.$$

Thus we determine the eigenvalues of the discrete Laplacian.

Consider first the continuous eigenvalue problem

$$\begin{cases} u_{xx} + u_{yy} = \lambda u, & (x, y) \in (0, 1)^2, \\ u = 0, & (x, y) \in \partial(0, 1)^2. \end{cases}$$

Let

$$u(x, y) = v(x) w(y).$$

Then

$$\frac{v''}{v} + \frac{w''}{w} = \lambda.$$

Hence,

$$v'' = \lambda_1 v, \qquad w'' = \lambda_2 w, \qquad \lambda = \lambda_1 + \lambda_2.$$

The boundary conditions give

$$v_p(x) = \sin(p\pi x), \qquad w_q(y) = \sin(q\pi y).$$

Therefore,

$$u^{(p,q)}(x, y) = \sin(p\pi x) \sin(q\pi y).$$

Motivated by the continuous eigenfunctions, define

$$U_{ij}^{(p,q)} = \sin(p\pi x_i)\sin(q\pi y_j).$$

Then

$$\Delta_h U_{ij}^{(p,q)} = \lambda_{p,q} U_{ij}^{(p,q)},$$

where

$$\lambda_{p,q} = \frac{2}{h^2}\left[\cos(p\pi h) + \cos(q\pi h) - 2\right].$$

Equivalently,

$$\lambda_{p,q} = -\frac{4}{h^2}\left[\sin^2\left(\frac{p\pi h}{2}\right) + \sin^2\left(\frac{q\pi h}{2}\right)\right].$$

Thus,

$$\lambda_{p,q} < 0.$$

The eigenvalue with the smallest absolute value is

$$\lambda_{1,1} = \frac{4}{h^2}\left(\cos(\pi h) - 1\right).$$

Therefore,

$$|\lambda_{1,1}| = \frac{4}{h^2}\left(1 - \cos(\pi h)\right).$$

For

$$0 < h \le \frac{1}{2},$$

the estimate in the notes gives

$$|\lambda_{1,1}| \ge 16.$$

Hence,

$$\boxed{\|A^{-1}\|_2 \le \frac{1}{16}}.$$

Thus the scheme is uniformly stable in the discrete L^2-norm.

If

$$u \in C^4([0,1]^2),$$

then

$$\|\tau\|_2 \le Ch^2.$$

Since

$$E = A^{-1}\tau,$$

we obtain

$$\|E\|_2 \le \|A^{-1}\|_2 \|\tau\|_2.$$

Therefore,

$$\boxed{\|E\|_2 \le \frac{1}{16}Ch^2}.$$

Hence the five-point scheme is second-order convergent in the discrete L^2-norm.

3.4 Maximum and Comparison Principles

Consider the general elliptic operator

$$Lu = a_{11}u_{xx} + 2a_{12}u_{xy} + a_{22}u_{yy} + b_1u_x + b_2u_y + cu.$$

It may also be written schematically as

$$Lu = \nabla \cdot (A\nabla u) + \mathbf{b} \cdot \nabla u + cu.$$

Assume that the matrix A is positive definite and

$$c \leq 0.$$

Theorem 3.4.1 (Maximum Principle)**.** *Suppose*

$$Lu = f$$

in Ω *and*

$$f \geq 0.$$

Then u *attains its maximum on the boundary. In particular, if*

$$u = g$$

on $\partial\Omega$*, then*

$$\sup_{\overline{\Omega}} u \leq \max_{\partial\Omega} g.$$

A physical example is

$$\Delta u = -\phi, \qquad \phi \geq 0.$$

The highest temperature is attained on the boundary.

The finite difference scheme satisfies an analogous property.

Theorem 3.4.2 (Discrete Maximum Principle)**.** *Suppose*

$$\Delta_h u_{ij} = f_{ij}$$

at all interior grid points and

$$u_{ij} = g_{ij}$$

on the discrete boundary. If

$$f_{ij} \geq 0,$$

then

$$\boxed{\max_{\Omega_h} u_{ij} \leq \max_{\partial\Omega_h} g_{ij}}.$$

Proof. Suppose the maximum is attained at an interior grid point (x_i, y_j). Then

$$u_{i-1,j} \leq u_{ij}, \qquad u_{i+1,j} \leq u_{ij},$$

and

$$u_{i,j-1} \leq u_{ij}, \qquad u_{i,j+1} \leq u_{ij}.$$

Hence,

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{ij} \leq 0.$$

Therefore,

$$\Delta_h u_{ij} \leq 0.$$

If

$$\Delta_h u_{ij} = f_{ij} \geq 0,$$

then equality must hold. Thus all four neighboring values must equal the same maximum. Repeating the argument propagates the maximum to the boundary, which proves the result. □

Corollary 3.4.3 (Discrete Comparison Principle)**.** *Suppose*

$$\Delta_h u_{ij} \geq \Delta_h v_{ij}$$

at all interior grid points and

$$u_{ij} = v_{ij}$$

on the discrete boundary. Then

$$u_{ij} \leq v_{ij}$$

throughout the grid.

Proof. Set

$$w_{ij} = u_{ij} - v_{ij}.$$

Then

$$\Delta_h w_{ij} \geq 0$$

in the interior and

$$w_{ij} = 0$$

on the boundary. By the discrete maximum principle,

$$w_{ij} \leq 0.$$

Hence,

$$u_{ij} \leq v_{ij}.$$

□

If

$$\Delta_h u_{ij} = 0,$$

then the numerical solution attains both its maximum and its minimum on the discrete boundary.

3.5 L^∞ Stability and Error

The discrete maximum principle also gives information about the inverse matrix.

For the homogeneous boundary problem,

$$AU = F.$$

If

$$F \geq 0$$

componentwise, the discrete maximum principle gives

$$U \leq 0.$$

Since

$$U = A^{-1}F,$$

we conclude that

$$\boxed{A^{-1} \leq 0}$$

componentwise.

Let

$$S = A^{-1}\mathbf{1},$$

where

$$\mathbf{1} = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^T.$$

Then

$$AS = \mathbf{1},$$

which is equivalent to

$$\begin{cases} \Delta_h S_{ij} = 1, & (x_i, y_j) \in \Omega_h, \\ S_{ij} = 0, & (x_i, y_j) \in \partial\Omega_h. \end{cases}$$

Since

$$A^{-1} \leq 0,$$

we have

$$S_{ij} \leq 0.$$

Construct the comparison function

$$w(x, y) = \frac{1}{4}\left(x - \frac{1}{2}\right)^2 + \frac{1}{4}\left(y - \frac{1}{2}\right)^2 - \frac{1}{8}.$$

Then

$$\Delta_h w_{ij} = 1.$$

Moreover,

$$w(x, y) \leq 0$$

on the square and

$$|w(x, y)| \leq \frac{1}{8}.$$

Using the comparison principle gives

$$w_{ij} \leq S_{ij} \leq 0.$$

Therefore,

$$|S_{ij}| \le \frac{1}{8}.$$

Since

$$A^{-1} \le 0,$$

we have

$$\|A^{-1}\|_\infty = \|A^{-1}\mathbf{1}\|_\infty = \|S\|_\infty.$$

Hence,

$$\boxed{\|A^{-1}\|_\infty \le \frac{1}{8}}.$$

Consider now

$$\begin{cases} \Delta_h u_{ij} = f_{ij}, & (x_i, y_j) \in \Omega_h, \\ u_{ij} = g_{ij}, & (x_i, y_j) \in \partial\Omega_h. \end{cases}$$

Write

$$U = V + W,$$

where

$$\begin{cases} \Delta_h V_{ij} = f_{ij}, & \Omega_h, \\ V_{ij} = 0, & \partial\Omega_h, \end{cases}$$

and

$$\begin{cases} \Delta_h W_{ij} = 0, & \Omega_h, \\ W_{ij} = g_{ij}, & \partial\Omega_h. \end{cases}$$

The inverse estimate gives

$$\|V\|_\infty \le \frac{1}{8}\|f\|_\infty.$$

The discrete maximum principle gives

$$\|W\|_\infty \le \|g\|_\infty.$$

Therefore,

$$\boxed{\|U\|_\infty \le \|g\|_\infty + \frac{1}{8}\|f\|_\infty}.$$

If

$$u \in C^4([0,1]^2),$$

the error satisfies

$$\begin{cases} \Delta_h e_{ij} = \tau_{ij}, & (x_i, y_j) \in \Omega_h, \\ e_{ij} = 0, & (x_i, y_j) \in \partial\Omega_h. \end{cases}$$

Thus,

$$\|E\|_\infty \le \frac{1}{8}\|\tau\|_\infty.$$

Since

$$\|\tau\|_\infty \le C_0 h^2,$$

we obtain

$$\boxed{\|E\|_\infty \le Ch^2}.$$

Therefore, the five-point finite difference method is second-order convergent in the discrete L^∞-norm.

3.6 Variable-Coefficient Elliptic Equations

Consider

$$\nabla \cdot (k \nabla u) = f.$$

In two dimensions,

$$(ku_x)_x + (ku_y)_y = f.$$

The natural discretization is obtained by approximating the fluxes at the midpoints of grid edges.

Define

$$k_{i+\frac{1}{2},j} = k\left(x_i + \frac{h}{2}, y_j\right),$$

and

$$k_{i,j+\frac{1}{2}} = k\left(x_i, y_j + \frac{h}{2}\right).$$

Approximate the right and left fluxes by

$$k_{i+\frac{1}{2},j} \frac{u_{i+1,j} - u_{ij}}{h},$$

and

$$k_{i-\frac{1}{2},j} \frac{u_{ij} - u_{i-1,j}}{h}.$$

Then

$$(ku_x)_x$$

is approximated by

$$\frac{1}{h^2}\left[k_{i+\frac{1}{2},j}(u_{i+1,j} - u_{ij}) - k_{i-\frac{1}{2},j}(u_{ij} - u_{i-1,j})\right].$$

Similarly,

$$(ku_y)_y$$

is approximated by

$$\frac{1}{h^2}\left[k_{i,j+\frac{1}{2}}(u_{i,j+1} - u_{ij}) - k_{i,j-\frac{1}{2}}(u_{ij} - u_{i,j-1})\right].$$

Therefore, the complete finite difference equation is

$$\begin{aligned}
&\frac{1}{h^2}\left[k_{i+\frac{1}{2},j}(u_{i+1,j} - u_{ij}) - k_{i-\frac{1}{2},j}(u_{ij} - u_{i-1,j})\right] \\
+&\frac{1}{h^2}\left[k_{i,j+\frac{1}{2}}(u_{i,j+1} - u_{ij}) - k_{i,j-\frac{1}{2}}(u_{ij} - u_{i,j-1})\right] = f_{ij}.
\end{aligned}$$

Equivalently,

$$\begin{aligned}
&k_{i-\frac{1}{2},j} u_{i-1,j} + k_{i+\frac{1}{2},j} u_{i+1,j} \\
&+k_{i,j-\frac{1}{2}} u_{i,j-1} + k_{i,j+\frac{1}{2}} u_{i,j+1} \\
-&\left(k_{i-\frac{1}{2},j} + k_{i+\frac{1}{2},j} + k_{i,j-\frac{1}{2}} + k_{i,j+\frac{1}{2}}\right) u_{ij} = h^2 f_{ij}.
\end{aligned}$$

The matrix remains symmetric. With the conventional sign choice

$$-\nabla \cdot (k \nabla u),$$

and under

$$k(x, y) > 0,$$

the resulting matrix is positive definite.

Remark 3.6.1. The discretization preserves the divergence form of the continuous operator. This is closely related to the conservative discretization from Chapter 2 and naturally leads to the finite volume viewpoint in the next chapter.

Chapter 4

Finite Volume Methods

4.1 Conservation Laws and Control Volumes

The finite volume method is based on integral conservation rather than pointwise differentiation.

Consider the one-dimensional conservation law

$$u_t + (f(u))_x = 0.$$

Here $u(x,t)$ denotes a conserved quantity and $f(u)$ denotes its flux.

Integrating over an interval $[a,b]$ gives

$$\int_a^b u_t(x,t)\,dx + \int_a^b (f(u))_x\,dx = 0.$$

Therefore,

$$\frac{d}{dt}\int_a^b u(x,t)\,dx = f(u(a,t)) - f(u(b,t)).$$

Thus the rate of change of the total amount of u inside the interval is determined entirely by the flux through its boundary.

Introduce a uniform mesh

$$x_j = jh.$$

Define the cell interfaces by

$$x_{j+\frac{1}{2}} = x_j + \frac{h}{2}.$$

The j-th control volume is

$$I_j = \left[x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}\right].$$

The cell average is

$$\overline{u}_j(t) = \frac{1}{h}\int_{I_j} u(x,t)\,dx.$$

Integrating the conservation law over I_j gives

$$\frac{d}{dt}\int_{I_j} u(x,t)\,dx + f\left(u(x_{j+\frac{1}{2}},t)\right) - f\left(u(x_{j-\frac{1}{2}},t)\right) = 0.$$

Dividing by h, we obtain

$$\frac{d}{dt}\overline{u}_j(t) = -\frac{1}{h}\left[f\left(u(x_{j+\frac{1}{2}},t)\right) - f\left(u(x_{j-\frac{1}{2}},t)\right)\right].$$

Remark 4.1.1. Finite difference methods are naturally based on point values and derivative approximations. Finite volume methods are naturally based on cell averages and flux balances.

4.2 Numerical Fluxes and the Finite Volume Scheme

The exact flux values at the cell interfaces are generally unknown. We therefore replace them by numerical fluxes.

Let

$$F_{j+\frac{1}{2}} = F(u_j, u_{j+1})$$

approximate the physical flux through the interface $x_{j+\frac{1}{2}}$.

Definition 4.2.1 (Consistent Numerical Flux)**.** A numerical flux $F(a,b)$ is called *consistent* with the physical flux f if

$$F(u,u) = f(u)$$

for every admissible state u.

The semi-discrete finite volume method is

$$\boxed{\frac{du_j}{dt} = -\frac{1}{h}\left(F_{j+\frac{1}{2}} - F_{j-\frac{1}{2}}\right).}$$

Here $u_j(t)$ approximates the cell average.

Introduce a time grid

$$t^n = n\tau.$$

Let

$$u_j^n$$

approximate the cell average at time t^n. Applying Forward Euler in time gives

$$u_j^{n+1} = u_j^n - \frac{\tau}{h}\left(F_{j+\frac{1}{2}}^n - F_{j-\frac{1}{2}}^n\right).$$

Define

$$\lambda = \frac{\tau}{h}.$$

Then

$$\boxed{u_j^{n+1} = u_j^n - \lambda\left(F_{j+\frac{1}{2}}^n - F_{j-\frac{1}{2}}^n\right).}$$

If

$$F_{j+\frac{1}{2}}^n = F(u_j^n, u_{j+1}^n),$$

then

$$u_j^{n+1} = u_j^n - \lambda\left[F(u_j^n, u_{j+1}^n) - F(u_{j-1}^n, u_j^n)\right].$$

4.3 Discrete Conservation

The main structural advantage of the finite volume method is that conservation is built directly into the scheme.

Consider cells

$$I_m, I_{m+1}, \ldots, I_n.$$

Summing the semi-discrete equations gives

$$\frac{d}{dt}\left(h\sum_{j=m}^{n} u_j\right) = -\sum_{j=m}^{n}\left(F_{j+\frac{1}{2}} - F_{j-\frac{1}{2}}\right).$$

The interior fluxes cancel:

$$\sum_{j=m}^{n}\left(F_{j+\frac{1}{2}} - F_{j-\frac{1}{2}}\right) = F_{n+\frac{1}{2}} - F_{m-\frac{1}{2}}.$$

Hence,

$$\boxed{\frac{d}{dt}\left(h\sum_{j=m}^{n} u_j\right) = F_{m-\frac{1}{2}} - F_{n+\frac{1}{2}}.}$$

Thus the change of the total discrete mass inside a group of cells is determined only by the flux through its outer boundary.

For the fully discrete scheme,

$$u_j^{n+1} = u_j^n - \lambda \left(F^n_{j+\frac{1}{2}} - F^n_{j-\frac{1}{2}} \right),$$

summing over j gives the same telescoping cancellation.

Remark 4.3.1. This telescoping property is the discrete counterpart of the divergence theorem. Internal fluxes cancel exactly between adjacent control volumes.

4.4 Upwind and Lax–Friedrichs Fluxes

Consider the linear advection equation

$$u_t + a u_x = 0.$$

Since

$$f(u) = au,$$

it can be written as

$$u_t + (au)_x = 0.$$

If

$$a > 0,$$

information propagates from left to right. The interface flux should therefore use the state on the left:

$$F(u_L, u_R) = a u_L.$$

Thus,

$$F_{j+\frac{1}{2}} = a u_j.$$

The finite volume scheme becomes

$$u_j^{n+1} = u_j^n - \frac{a\tau}{h} \left(u_j^n - u_{j-1}^n \right).$$

If

$$a < 0,$$

information propagates from right to left, so

$$F(u_L, u_R) = a u_R.$$

Then

$$u_j^{n+1} = u_j^n - \frac{a\tau}{h} \left(u_{j+1}^n - u_j^n \right).$$

The upwind flux can be written compactly as

$$\boxed{F(u_L, u_R) = a^+ u_L + a^- u_R},$$

where

$$a^+ = \max(a, 0), \qquad a^- = \min(a, 0).$$

For linear advection, the finite volume upwind scheme and the finite difference upwind scheme have the same algebraic form. Their interpretations are different: the finite

difference method approximates u_x, while the finite volume method approximates the flux across cell interfaces.

Another important choice is the Lax–Friedrichs flux:

$$F_{\mathrm{LF}}(u_L, u_R) = \frac{1}{2}\left[f(u_L) + f(u_R)\right] - \frac{\alpha}{2}\left(u_R - u_L\right).$$

The parameter $\alpha > 0$ is chosen sufficiently large. For a scalar conservation law, a standard choice is

$$\alpha \geq \max |f'(u)|$$

over the relevant range of states.

The first term is a centered approximation of the physical flux, while the second term adds numerical dissipation.

For

$$f(u) = au,$$

choose

$$\alpha = |a|.$$

Then

$$F_{\mathrm{LF}}(u_L, u_R) = \frac{a}{2}\left(u_L + u_R\right) - \frac{|a|}{2}\left(u_R - u_L\right).$$

If $a > 0$, this reduces to

$$F_{\mathrm{LF}}(u_L, u_R) = au_L.$$

If $a < 0$, it reduces to

$$F_{\mathrm{LF}}(u_L, u_R) = au_R.$$

Thus, for constant-coefficient linear advection, the Lax–Friedrichs flux with $\alpha = |a|$ coincides with the upwind flux.

4.5 Consistency, Reconstruction, and Stability

Suppose $u(x,t)$ is smooth. The exact cell average satisfies

$$\frac{d}{dt}\overline{u}_j = -\frac{1}{h}\left[f\left(u(x_{j+\frac{1}{2}}, t)\right) - f\left(u(x_{j-\frac{1}{2}}, t)\right)\right].$$

A consistent numerical flux satisfies

$$F(u, u) = f(u).$$

If neighboring cell averages satisfy

$$u_j = u(x_j, t) + O(h^2),$$

then for a sufficiently smooth numerical flux,

$$F(u_j, u_{j+1}) = f\left(u(x_{j+\frac{1}{2}}, t)\right) + O(h).$$

Thus a first-order flux generally gives a first-order spatial method.

The simplest reconstruction assumes that the solution is constant inside each cell:

$$u_h(x,t) = u_j(t), \qquad x \in I_j.$$

At the interface $x_{j+\frac{1}{2}}$, the left and right states are

$$u^-_{j+\frac{1}{2}} = u_j, \qquad u^+_{j+\frac{1}{2}} = u_{j+1}.$$

The numerical flux is then

$$F_{j+\frac{1}{2}} = F\left(u^-_{j+\frac{1}{2}}, u^+_{j+\frac{1}{2}}\right).$$

Piecewise constant reconstruction therefore leads naturally to first-order finite volume methods.

Higher-order finite volume schemes use more accurate reconstruction, for example piecewise linear reconstruction.

For the linear advection equation with $a > 0$, the upwind scheme is

$$u^{n+1}_j = u^n_j - \nu\left(u^n_j - u^n_{j-1}\right),$$

where

$$\nu = \frac{a\tau}{h}.$$

Equivalently,

$$u^{n+1}_j = (1-\nu)u^n_j + \nu u^n_{j-1}.$$

If

$$0 \le \nu \le 1,$$

the new value is a convex combination of two values from the previous time level. Therefore,

$$|u^{n+1}_j| \le \max\left\{|u^n_j|, |u^n_{j-1}|\right\}.$$

Hence,

$$\|U^{n+1}\|_\infty \le \|U^n\|_\infty.$$

Thus the method is strongly stable under

$$\boxed{0 \le \frac{a\tau}{h} \le 1}.$$

For a general nonlinear conservation law, the corresponding restriction has the form

$$\frac{\tau}{h}\max|f'(u)| \le C,$$

where C depends on the numerical flux and time discretization.

4.6 Finite Volume Methods for Elliptic Problems

Finite volume methods can also be applied to elliptic equations.

Consider

$$-\nabla\cdot(k\nabla u) = f.$$

Integrating over a control volume V_i gives

$$-\int_{V_i}\nabla\cdot(k\nabla u)\,dx = \int_{V_i} f\,dx.$$

By the divergence theorem,

$$\boxed{-\int_{\partial V_i} k\nabla u\cdot n\,ds = \int_{V_i} f\,dx}.$$

Thus the differential equation becomes a balance of normal fluxes through the boundary of the control volume.

In one dimension, consider

$$-\Big(k(x)u'(x)\Big)' = f(x).$$

Integrating over

$$I_j = \left[x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}\right]$$

gives

$$-k_{j+\frac{1}{2}}u'\left(x_{j+\frac{1}{2}}\right) + k_{j-\frac{1}{2}}u'\left(x_{j-\frac{1}{2}}\right) = \int_{I_j} f(x)\,dx.$$

Approximate

$$u'\left(x_{j+\frac{1}{2}}\right) \approx \frac{u_{j+1}-u_j}{h},$$

and

$$u'\left(x_{j-\frac{1}{2}}\right) \approx \frac{u_j-u_{j-1}}{h}.$$

Then

$$-k_{j+\frac{1}{2}}\frac{u_{j+1}-u_j}{h} + k_{j-\frac{1}{2}}\frac{u_j-u_{j-1}}{h} \approx \int_{I_j} f(x)\,dx.$$

Dividing by h,

$$-\frac{1}{h^2}\left[k_{j+\frac{1}{2}}(u_{j+1}-u_j) - k_{j-\frac{1}{2}}(u_j-u_{j-1})\right] \approx \frac{1}{h}\int_{I_j} f(x)\,dx.$$

This is closely related to the conservative finite difference discretization derived in Chapter 2. On a uniform mesh, the two methods may produce the same algebraic equations, even though their derivations are different:

Finite Difference :	approximate the differential operator,
Finite Volume :	approximate the integral flux balance.

4.7 Boundary Conditions and Multidimensional Extension

Boundary conditions are naturally imposed through the fluxes at boundary faces.

For a Dirichlet condition

$$u(a) = \alpha,$$

the prescribed value is used to construct the numerical flux at the boundary interface.

For a Neumann condition

$$-k(a)u'(a) = q_a,$$

the boundary flux is already known and can be inserted directly into the balance equation.

This makes flux boundary conditions particularly natural in the finite volume framework.

For a two-dimensional conservation law,

$$u_t + \nabla \cdot F(u) = 0,$$

write

$$F(u) = \begin{bmatrix} f(u) \\ g(u) \end{bmatrix}.$$

Let

$$C_{ij} = \left[x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}\right] \times \left[y_{j-\frac{1}{2}}, y_{j+\frac{1}{2}}\right].$$

Integrating over C_{ij} gives

$$\frac{d}{dt}\int_{C_{ij}} u\,dx\,dy + \int_{\partial C_{ij}} F(u)\cdot n\,ds = 0.$$

For a rectangular grid with

$$\Delta x = h_x, \qquad \Delta y = h_y,$$

the semi-discrete method becomes

$$\frac{du_{ij}}{dt} = -\frac{1}{h_x}\left(F_{i+\frac{1}{2},j} - F_{i-\frac{1}{2},j}\right)$$

$$-\frac{1}{h_y}\left(G_{i,j+\frac{1}{2}} - G_{i,j-\frac{1}{2}}\right).$$

Thus each cell interacts with its neighbors only through fluxes across its faces.

If two neighboring cells share an interface, the flux leaving one cell enters the other with the opposite sign. Therefore, internal fluxes cancel when equations are summed over a larger region.

This local conservation property is one of the principal advantages of the finite volume method.

Remark 4.7.1. For smooth solutions, finite volume methods can be analyzed through the same consistency–stability–convergence framework used for finite difference methods. For nonlinear conservation laws, discontinuities may develop even from smooth initial data, so the concept of solution and the role of conservative discretization become more subtle. These issues will be revisited in Chapter 11.

Chapter 5

Weak Formulations and Galerkin Methods

5.1 Weak Derivatives and Sobolev Spaces

Let

$$\Omega \subset \mathbb{R}^d,$$

and let

$$\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$$

be a multi-index. Its order is

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d.$$

For a sufficiently smooth function u, define

$$D^\alpha u = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_d^{\alpha_d}}.$$

Define

$$C_0^\infty(\Omega) = \{\phi \in C^\infty(\Omega) : \operatorname{supp} \phi \subset \Omega\}.$$

Functions in $C_0^\infty(\Omega)$ are called test functions.

Definition 5.1.1 (Weak Derivative)**.** Let

$$f \in L^1_{\mathrm{loc}}(\Omega).$$

A function $g \in L^1_{\mathrm{loc}}(\Omega)$ is called the weak derivative $D^\alpha f$ if

$$\int_\Omega g(x)\phi(x)\,dx = (-1)^{|\alpha|} \int_\Omega f(x) D^\alpha \phi(x)\,dx$$

for every

$$\phi \in C_0^\infty(\Omega).$$

If such a function g exists, we write

$$D_w^\alpha f = g.$$

The definition is motivated by repeated integration by parts. Since the test function has compact support in Ω, the boundary terms vanish.

Consider the example

$$\Omega = (-1, 1), \qquad f(x) = 1 - |x|.$$

The function is not classically differentiable at $x = 0$. Define

$$g(x) = \begin{cases} 1, & x < 0, \\ -1, & x > 0. \end{cases}$$

For every

$$\phi \in C_0^\infty(-1, 1),$$

we have

$$\int_{-1}^{1} f(x)\phi'(x)\,dx = \int_{-1}^{0} (1 + x)\phi'(x)\,dx + \int_{0}^{1} (1 - x)\phi'(x)\,dx.$$

Integrating by parts gives

$$\int_{-1}^{1} f(x)\phi'(x)\,dx = -\int_{-1}^{0} \phi(x)\,dx + \int_{0}^{1} \phi(x)\,dx.$$

Therefore,

$$\int_{-1}^{1} f(x)\phi'(x)\,dx = -\int_{-1}^{1} g(x)\phi(x)\,dx.$$

Hence,

$$D_w f = g.$$

Remark 5.1.2. Weak differentiability is weaker than classical differentiability. A function may fail to be differentiable at some points while still possessing a weak derivative.

Weak derivatives lead naturally to Sobolev spaces.

Definition 5.1.3 (Sobolev Space $H^1(\Omega)$)**.** The space $H^1(\Omega)$ is defined by

$$H^1(\Omega) = \left\{ u \in L^2(\Omega) : D_j u \in L^2(\Omega),\ 1 \le j \le d \right\}.$$

Its norm is

$$\|u\|^2_{H^1(\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\nabla u\|^2_{L^2(\Omega)}.$$

For homogeneous Dirichlet boundary conditions, we use

$$H^1_0(\Omega).$$

In these notes, this space may be viewed as

$$H^1_0(\Omega) = \left\{ u \in H^1(\Omega) : u|_{\partial\Omega} = 0 \right\}.$$

More precisely, $H^1_0(\Omega)$ is the closure of $C^\infty_0(\Omega)$ in the H^1-norm.

5.2 Weak Formulation of the Poisson Equation

Consider

$$\begin{cases} -\Delta u = f, & x \in \Omega, \\ u = 0, & x \in \partial\Omega. \end{cases}$$

A strong solution requires sufficient differentiability so that the equation holds pointwise. For example, one may require

$$u \in C^2(\Omega).$$

To obtain the weak formulation, multiply the equation by

$$v \in H^1_0(\Omega)$$

and integrate:

$$-\int_\Omega \Delta u\, v\,dx = \int_\Omega f v\,dx.$$

Using integration by parts,

$$-\int_\Omega \Delta u\, v\,dx = \int_\Omega \nabla u \cdot \nabla v\,dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} v\,ds.$$

Since

$$v = 0$$

on $\partial\Omega$, the boundary term vanishes. Therefore,

$$\boxed{\int_\Omega \nabla u \cdot \nabla v\,dx = \int_\Omega f v\,dx}.$$

Definition 5.2.1 (Weak Solution)**.** A function

$$u \in H_0^1(\Omega)$$

is called a weak solution if

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx$$

for every

$$v \in H_0^1(\Omega).$$

The weak formulation requires only first weak derivatives of u, rather than second classical derivatives.

5.3 Lax–Milgram Theory and Well-Posedness

The weak problem can be written abstractly using a bilinear form.

Definition 5.3.1 (Bilinear Form)**.** A mapping

$$a(\cdot, \cdot) : X \times X \longrightarrow \mathbb{R}$$

is called a bilinear form if it is linear in each variable.

Definition 5.3.2 (Inner Product and Hilbert Space)**.** An inner product $(\cdot, \cdot)$ is a symmetric bilinear form satisfying

$$(u, u) > 0$$

for every nonzero $u \in X$. The associated norm is

$$\|u\|_X = \sqrt{(u, u)}.$$

A Hilbert space is a complete inner-product space.

Let X be a normed linear space.

Definition 5.3.3 (Dual Space)**.** The dual space X' consists of all bounded linear functionals

$$F : X \longrightarrow \mathbb{R}.$$

That is, $F \in X'$ if there exists $C > 0$ such that

$$|F(v)| \leq C\|v\|_X$$

for every $v \in X$.

The dual norm is

$$\|F\|_{X'} = \sup_{v \neq 0} \frac{|F(v)|}{\|v\|_X}.$$

Theorem 5.3.4 (Lax–Milgram Lemma)**.** *Let X be a Hilbert space and let*

$$a : X \times X \longrightarrow \mathbb{R}$$

be a bilinear form. Assume that there exists $C > 0$ such that

$$|a(u, v)| \leq C\|u\|_X \|v\|_X$$

for all $u, v \in X$, and there exists $\alpha > 0$ such that

$$a(u,u) \geq \alpha \|u\|_X^2$$

for all $u \in X$. Then, for every $F \in X'$, there exists a unique $u \in X$ such that

$$a(u,v) = F(v)$$

for every $v \in X$. Moreover,

$$\boxed{\|u\|_X \leq \frac{1}{\alpha}\|F\|_{X'}}.$$

The first condition is called continuity and the second is called coercivity.

For the Poisson problem, set

$$X = H_0^1(\Omega),$$

and define

$$a(u,v) = \int_\Omega \nabla u \cdot \nabla v \, dx,$$

$$F(v) = \int_\Omega f v \, dx.$$

The continuity of a follows from Cauchy–Schwarz:

$$|a(u,v)| \leq \|\nabla u\|_{L^2} \|\nabla v\|_{L^2}.$$

Theorem 5.3.5 (Poincaré Inequality)**.** *There exists a constant $C_P > 0$, depending only on Ω and d, such that*

$$\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)}$$

for every

$$u \in H_0^1(\Omega).$$

Using the Poincaré inequality,

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2$$

satisfies

$$\|u\|_{H^1}^2 \leq (C_P^2 + 1)\|\nabla u\|_{L^2}^2.$$

Therefore,

$$a(u,u) = \|\nabla u\|_{L^2}^2 \geq \frac{1}{C_P^2 + 1}\|u\|_{H^1}^2.$$

Thus a is coercive. By the Lax–Milgram lemma, there exists a unique

$$u \in H_0^1(\Omega)$$

such that

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx$$

for every

$$v \in H_0^1(\Omega).$$

5.4 Galerkin Approximation and Céa's Lemma

The weak problem is still infinite-dimensional. The Galerkin method replaces the space X by a finite-dimensional subspace

$$X_N \subset X.$$

The Galerkin approximation seeks

$$u_N \in X_N$$

such that

$$a(u_N, v_N) = F(v_N)$$

for every

$$v_N \in X_N.$$

By the Lax–Milgram lemma applied to X_N, the Galerkin solution exists uniquely.

Let

$$X_N = \operatorname{span}\{\phi_1, \phi_2, \ldots, \phi_N\}.$$

Write

$$u_N = \sum_{k=1}^{N} c_k \phi_k.$$

Taking

$$v_N = \phi_j$$

gives

$$\sum_{k=1}^{N} c_k a(\phi_k, \phi_j) = F(\phi_j).$$

Define

$$A_{jk} = a(\phi_k, \phi_j), \qquad f_j = F(\phi_j).$$

Then

$$\boxed{A\mathbf{c} = \mathbf{f}}.$$

For every nonzero coefficient vector $\mathbf{c}$,

$$\mathbf{c}^T A \mathbf{c} = a\left(\sum_{k=1}^{N} c_k \phi_k, \sum_{j=1}^{N} c_j \phi_j\right).$$

By coercivity,

$$\mathbf{c}^T A \mathbf{c} \geq \alpha \left\| \sum_{k=1}^{N} c_k \phi_k \right\|_X^2 > 0.$$

Thus A is positive definite and invertible.

The exact weak solution satisfies

$$a(u, v_N) = F(v_N),$$

while the Galerkin solution satisfies

$$a(u_N, v_N) = F(v_N).$$

Subtracting gives

$$\boxed{a(u - u_N, v_N) = 0}$$

for every

$$v_N \in X_N.$$

This is called *Galerkin orthogonality.*

Theorem 5.4.1 (Céa's Lemma)**.** *Suppose*

$$|a(u,v)| \le C\|u\|_X\|v\|_X$$

and

$$a(u,u) \ge \alpha\|u\|_X^2.$$

Then

$$\boxed{\|u-u_N\|_X \le \frac{C}{\alpha}\inf_{v_N\in X_N}\|u-v_N\|_X}.$$

Proof. Let

$$e_N = u - u_N.$$

By coercivity,

$$\alpha\|e_N\|_X^2 \le a(e_N,e_N).$$

For any $v_N \in X_N$,

$$e_N = u - v_N + v_N - u_N.$$

Therefore,

$$a(e_N,e_N) = a(e_N,u-v_N) + a(e_N,v_N-u_N).$$

Since

$$v_N - u_N \in X_N,$$

Galerkin orthogonality gives

$$a(e_N,v_N-u_N) = 0.$$

Hence,

$$\alpha\|e_N\|_X^2 \le a(e_N,u-v_N).$$

By continuity,

$$\alpha\|e_N\|_X^2 \le C\|e_N\|_X\|u-v_N\|_X.$$

Therefore,

$$\|e_N\|_X \le \frac{C}{\alpha}\|u-v_N\|_X.$$

Taking the infimum over $v_N \in X_N$ proves the result. □

5.5 Energy Norm and L^2 Error Estimate

If the bilinear form is symmetric and positive definite, define the energy norm by

$$\|u\|_E = \sqrt{a(u,u)}.$$

For any $v_N \in X_N$,

$$u - v_N = (u-u_N) + (u_N - v_N).$$

Galerkin orthogonality gives

$$a(u-u_N, u_N - v_N) = 0.$$

Therefore,

$$\|u - v_N\|_E^2 = \|u - u_N\|_E^2 + \|u_N - v_N\|_E^2.$$

Hence,

$$\boxed{\|u - u_N\|_E = \min_{v_N \in X_N} \|u - v_N\|_E}.$$

Thus the Galerkin solution is the best approximation in the energy norm.

We now estimate the error in the L^2-norm. Let

$$e = u - u_N.$$

For the one-dimensional Poisson problem, let w solve

$$\begin{cases} -w'' = e, & x \in (0,1), \\ w(0) = w(1) = 0. \end{cases}$$

Then

$$\|e\|_{L^2}^2 = (e, -w'').$$

Integration by parts gives

$$\|e\|_{L^2}^2 = a(e, w).$$

By Galerkin orthogonality,

$$a(e, v_N) = 0$$

for every $v_N \in X_N$. Therefore,

$$\|e\|_{L^2}^2 = a(e, w - v_N).$$

Using Cauchy–Schwarz in the energy norm,

$$\|e\|_{L^2}^2 \le \|e\|_E \|w - v_N\|_E.$$

If

$$\inf_{v_N \in X_N} \|w - v_N\|_E \le Ch\|w''\|_{L^2},$$

then

$$\|e\|_{L^2}^2 \le Ch\|e\|_E\|w''\|_{L^2}.$$

Since

$$-w'' = e,$$

we have

$$\|w''\|_{L^2} = \|e\|_{L^2}.$$

Therefore,

$$\|e\|_{L^2} \le Ch\|e\|_E.$$

If

$$\|e\|_E \le Ch\|u''\|_{L^2},$$

then

$$\boxed{\|u - u_N\|_{L^2} \le Ch^2\|u''\|_{L^2}}.$$

Remark 5.5.1. The auxiliary problem for w converts an energy-norm estimate into a stronger L^2-estimate. This is a standard duality argument.

5.6 Fourier Spectral Galerkin Method

The Galerkin framework is not restricted to piecewise polynomial spaces. Consider the periodic Poisson equation

$$\Delta u = f$$

on

$$\Omega = [0, 2\pi]^d.$$

To guarantee solvability, impose the compatibility condition

$$\int_\Omega f(x)\, dx = 0.$$

To obtain uniqueness, impose

$$\int_\Omega u(x)\, dx = 0.$$

Define

$$X = \left\{ u \in H^1_{\text{per}}(\Omega) : \int_\Omega u(x)\, dx = 0 \right\}.$$

Expand u in a Fourier series:

$$u(x) = \sum_{k \in \mathbb{Z}^d} \widehat{u}_k e^{ik\cdot x}.$$

Since the mean is zero,

$$\widehat{u}_0 = 0.$$

The periodic Sobolev norm is

$$\|u\|^2_{H^s_{\text{per}}} = \sum_{k \in \mathbb{Z}^d} (1 + |k|^2)^s |\widehat{u}_k|^2.$$

Define the finite-dimensional space

$$X_N = \text{span}\left\{ e^{ik\cdot x} : 0 < |k| \le N \right\}.$$

Write

$$u_N(x) = \sum_{0<|k|\le N} c_k e^{ik\cdot x}.$$

The Fourier modes are orthogonal, so the Galerkin matrix is diagonal.

For the Poisson equation, testing against a Fourier mode gives

$$|j|^2 c_j = \widehat{f}_j$$

up to the sign convention used for the Laplacian. Hence,

$$c_j = \frac{\widehat{f}_j}{|j|^2}.$$

Define the Fourier projection

$$\Pi_N : X \longrightarrow X_N$$

by

$$\Pi_N u(x) = \sum_{0<|k|\le N} \widehat{u}_k e^{ik\cdot x}.$$

Then

$$u - \Pi_N u = \sum_{|k|>N} \widehat{u}_k e^{ik\cdot x}.$$

Therefore,

$$\|u - \Pi_N u\|_{H^r}^2 = \sum_{|k|>N} (1 + |k|^2)^r |\widehat{u}_k|^2.$$

If

$$u \in H^s, \qquad s \geq r,$$

then

$$(1 + |k|^2)^r \leq C N^{-2(s-r)} (1 + |k|^2)^s$$

for $|k| > N$. Hence,

$$\boxed{\|u - \Pi_N u\|_{H^r} \leq C N^{-(s-r)} \|u\|_{H^s}}.$$

Thus the approximation error tends to zero as

$$N \longrightarrow \infty.$$

The projection $\Pi_N u$ also satisfies the Galerkin equations:

$$a(\Pi_N u, v_N) = F(v_N)$$

for every

$$v_N \in X_N.$$

Therefore, for this periodic constant-coefficient problem, the Fourier projection and the Fourier Galerkin solution coincide.

Chapter 6

Finite Element Methods and Linear Solvers

6.1 Boundary Conditions in Variational Problems

Consider the two-point boundary value problem

$$\begin{cases} -u''(x) = f(x), & x \in (0,1), \\ u(0) = 0, \qquad u'(1) = 0. \end{cases}$$

Set

$$X = \left\{ u \in H^1(0,1) : u(0) = 0 \right\},$$

and define

$$a(u,v) = \int_0^1 u'(x)v'(x)\,dx.$$

The variational problem is to find $u \in X$ such that

$$\boxed{a(u,v) = (f,v)}$$

for every $v \in X$, where

$$(f,v) = \int_0^1 f(x)v(x)\,dx.$$

We now verify that this variational problem recovers the original boundary value problem.

If u is a sufficiently smooth solution of the differential equation, then

$$-u'' = f.$$

Multiplying by $v \in X$ and integrating gives

$$\int_0^1 f(x)v(x)\,dx = -\int_0^1 u''(x)v(x)\,dx.$$

Integration by parts gives

$$-\int_0^1 u''v\,dx = \int_0^1 u'v'\,dx - u'(1)v(1) + u'(0)v(0).$$

Since

$$v(0) = 0$$

and

$$u'(1) = 0,$$

we obtain

$$(f,v) = a(u,v).$$

Conversely, suppose

$$a(u,v) = (f,v)$$

for every $v \in X$. Then

$$\int_0^1 u'v'\,dx = \int_0^1 fv\,dx.$$

For smooth u, integration by parts gives

$$\int_0^1 u'v'\,dx = -\int_0^1 u''v\,dx + u'(1)v(1),$$

because $v(0) = 0$. Therefore,

$$\int_0^1 (-u'' - f)v\,dx + u'(1)v(1) = 0.$$

Choose test functions with

$$v(1) = 0.$$

Then

$$\int_0^1 (-u'' - f)v\,dx = 0$$

for all such test functions, which implies

$$-u'' = f$$

in $(0, 1)$. Substituting this relation back into the variational identity gives

$$u'(1)v(1) = 0$$

for every $v \in X$. Since $v(1)$ can be arbitrary,

$$u'(1) = 0.$$

Thus the variational formulation contains both the differential equation and the Neumann boundary condition.

The two common types of boundary conditions play different roles in the variational formulation:

Boundary condition	Variational name	Standard name
$u = 0$	essential / explicit	Dirichlet
$u' = 0$	natural / implicit	Neumann

A Dirichlet condition is imposed directly in the function space. A Neumann condition appears naturally after integration by parts.

6.2 Piecewise Linear Finite Element Spaces

Let

$$0 = x_0 < x_1 < \cdots < x_N = 1$$

be a partition of the interval, and define

$$h_i = x_i - x_{i-1}, \qquad i = 1, \ldots, N.$$

Let

$$h = \max_{1 \le i \le N} h_i.$$

For each node x_i, define the piecewise linear hat function

$$\phi_i(x) = \begin{cases} \dfrac{x - x_{i-1}}{x_i - x_{i-1}}, & x \in [x_{i-1}, x_i], \\ \dfrac{x_{i+1} - x}{x_{i+1} - x_i}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases}$$

The basis functions satisfy

$$\phi_i(x_j) = \delta_{ij}.$$

Their supports are local:

$$\operatorname{supp} \phi_i = [x_{i-1}, x_{i+1}].$$

For a homogeneous Dirichlet problem, the finite element space is

$$X_h = \left\{ v_h \in C^0[0,1] : v_h|_{[x_{i-1},x_i]} \text{ is linear}, \quad v_h(0) = v_h(1) = 0 \right\}.$$

Hence,

$$X_h = \operatorname{span} \{\phi_1, \ldots, \phi_{N-1}\}.$$

For the mixed problem

$$u(0) = 0, \qquad u'(1) = 0,$$

the value at $x = 1$ is not prescribed explicitly, so the basis function associated with the right endpoint is retained.

Any finite element function can be written as

$$u_h(x) = \sum_i c_i \phi_i(x).$$

Because each ϕ_i has local support, only neighboring basis functions interact. This locality is the source of sparsity in finite element matrices.

6.3 Interpolation and Error Estimates

Define the interpolation operator

$$I_h : C^0[0,1] \longrightarrow X_h$$

by

$$(I_h u)(x_i) = u(x_i)$$

at every unconstrained finite element node. Equivalently,

$$I_h u = \sum_i u(x_i) \phi_i.$$

For every $v_h \in X_h$,

$$I_h v_h = v_h.$$

For the one-dimensional Poisson problem, define the energy norm by

$$\|v\|_E = \left(\int_0^1 |v'(x)|^2 \, dx \right)^{1/2}.$$

Theorem 6.3.1 (Piecewise Linear Interpolation Error)**.** *Suppose*

$$u \in H^2(0,1).$$

Then

$$\boxed{\|u - I_h u\|_E \le Ch \|u''\|_{L^2(0,1)}}.$$

Proof. Define

$$e(x) = u(x) - I_h u(x).$$

On each element

$$[x_{i-1}, x_i],$$

the interpolant $I_h u$ is linear. Therefore,

$$e''(x) = u''(x).$$

It is enough to prove

$$\int_{x_{i-1}}^{x_i} |e'(x)|^2 \, dx \le C h_i^2 \int_{x_{i-1}}^{x_i} |u''(x)|^2 \, dx.$$

Map $[x_{i-1}, x_i]$ to $[0,1]$ by

$$x = x_{i-1} + h_i t$$

and define

$$\widetilde{e}(t) = e(x_{i-1} + h_i t).$$

Since the interpolation is exact at the endpoints,

$$\widetilde{e}(0) = \widetilde{e}(1) = 0.$$

By Rolle's theorem, there exists

$$\xi \in (0,1)$$

such that

$$\widetilde{e}'(\xi) = 0.$$

Let

$$w(t) = \widetilde{e}'(t).$$

Then

$$w(\xi) = 0.$$

For any $t \in [0,1]$,

$$w(t) = \int_{\xi}^{t} w'(s) \, ds.$$

By the Cauchy–Schwarz inequality,

$$|w(t)|^2 \le \int_0^1 |w'(s)|^2 \, ds.$$

Integrating with respect to t gives

$$\int_0^1 |w(t)|^2 \, dt \le \int_0^1 |w'(t)|^2 \, dt.$$

Transforming back to the original element gives

$$\int_{x_{i-1}}^{x_i} |e'(x)|^2 \, dx \le h_i^2 \int_{x_{i-1}}^{x_i} |u''(x)|^2 \, dx.$$

Since

$$h_i \le h,$$

summing over all elements yields

$$\int_0^1 |e'(x)|^2\,dx \le h^2 \int_0^1 |u''(x)|^2\,dx.$$

Therefore,

$$\|u - I_h u\|_E \le h\|u''\|_{L^2(0,1)}.$$

□

Combining this interpolation estimate with Céa's lemma gives

$$\|u - u_h\|_E \le C \inf_{v_h \in X_h} \|u - v_h\|_E.$$

Choosing

$$v_h = I_h u$$

gives

$$\boxed{\|u - u_h\|_E \le Ch\|u''\|_{L^2}}.$$

Using the duality argument from Chapter 5 further gives

$$\boxed{\|u - u_h\|_{L^2} \le Ch^2\|u''\|_{L^2}}.$$

Thus piecewise linear finite elements are first-order accurate in the energy norm and second-order accurate in the L^2-norm.

6.4 Finite Element Linear Systems

Consider

$$\begin{cases} -u''(x) = f(x), & x \in (0,1), \\ u(0) = u(1) = 0. \end{cases}$$

The finite element approximation seeks

$$u_h \in X_h$$

such that

$$a(u_h, v_h) = (f, v_h)$$

for every $v_h \in X_h$, where

$$a(u_h, v_h) = \int_0^1 u_h'(x) v_h'(x)\,dx.$$

Write

$$u_h(x) = \sum_{i=1}^{N-1} c_i \phi_i(x).$$

Taking $v_h = \phi_j$ gives

$$\sum_{i=1}^{N-1} c_i a(\phi_i, \phi_j) = (f, \phi_j).$$

Therefore,

$$\boxed{A\mathbf{c} = \mathbf{f}},$$

where

$$A_{ij} = a(\phi_i, \phi_j) = \int_0^1 \phi_i'(x)\phi_j'(x)\,dx$$

is called the stiffness matrix.

Since

$$\phi_i'(x) = \begin{cases} \dfrac{1}{h_i}, & x \in [x_{i-1}, x_i], \\ -\dfrac{1}{h_{i+1}}, & x \in [x_i, x_{i+1}], \\ 0, & \text{otherwise}, \end{cases}$$

we obtain

$$A_{ij} = \begin{cases} \dfrac{1}{h_i} + \dfrac{1}{h_{i+1}}, & j = i, \\ -\dfrac{1}{h_i}, & j = i-1, \\ -\dfrac{1}{h_{i+1}}, & j = i+1, \\ 0, & |i-j| > 1. \end{cases}$$

Thus A is symmetric, sparse, and tridiagonal. The notes also record that it is positive definite and strictly diagonally dominant.

For every nonzero coefficient vector $\mathbf{c}$,

$$\mathbf{c}^T A \mathbf{c} = \int_0^1 \left| \sum_i c_i \phi_i'(x) \right|^2 dx > 0.$$

Hence A is symmetric positive definite.

The load vector is

$$f_i = \int_0^1 f(x)\phi_i(x)\,dx.$$

Using the trapezoidal approximation recorded in the notes,

$$f_i \approx \frac{1}{2}\,(h_i + h_{i+1})\,f(x_i).$$

Therefore, the i-th finite element equation becomes

$$-\frac{1}{h_i}c_{i-1} + \left(\frac{1}{h_i} + \frac{1}{h_{i+1}}\right)c_i - \frac{1}{h_{i+1}}c_{i+1} = f_i.$$

On a uniform mesh,

$$h_i = h,$$

so

$$-\frac{1}{h}c_{i-1} + \frac{2}{h}c_i - \frac{1}{h}c_{i+1} = hf(x_i).$$

Dividing by h gives

$$\boxed{\frac{-c_{i-1} + 2c_i - c_{i+1}}{h^2} = f(x_i)}.$$

Thus, with this quadrature rule, the piecewise linear finite element method produces the same algebraic three-point stencil as the centered finite difference method on a uniform one-dimensional mesh.

6.5 Fast Direct Solvers

After discretization, we need to solve a linear system

$$AU = F.$$

If A can be diagonalized as

$$A = X\Lambda X^{-1},$$

then

$$U = X\Lambda^{-1}X^{-1}F.$$

When multiplication by X and X^{-1} can be computed by a fast transform, the system can be solved efficiently.

For the one-dimensional Dirichlet problem, the natural eigenvectors are discrete sine functions. Define the discrete sine transform by

$$y(k) = \sum_{n=1}^{N} x(n) \sin\left(\frac{\pi nk}{N+1}\right), \qquad k = 1, \ldots, N.$$

In matrix form,

$$y = Xx,$$

where

$$X_{kn} = \sin\left(\frac{\pi nk}{N+1}\right).$$

The inverse transform is

$$x(n) = \frac{2}{N+1} \sum_{k=1}^{N} y(k) \sin\left(\frac{\pi nk}{N+1}\right).$$

Fast sine transforms have computational cost

$$O(N \log N).$$

For periodic problems, use the discrete Fourier transform:

$$\widehat{x}(k) = \sum_{j=1}^{N} x(j) \exp\left(-\frac{2\pi i}{N}(j-1)(k-1)\right).$$

The inverse transform is

$$x(j) = \frac{1}{N} \sum_{k=1}^{N} \widehat{x}(k) \exp\left(\frac{2\pi i}{N}(j-1)(k-1)\right).$$

Consider the periodic finite difference equation

$$\frac{u_{j-1} - 2u_j + u_{j+1}}{h^2} = f_j.$$

Applying the discrete Fourier transform gives

$$\lambda_k \widehat{U}(k) = \widehat{F}(k),$$

where

$$\lambda_k = \frac{2\cos\left(\frac{2\pi(k-1)}{N}\right) - 2}{h^2}.$$

For the zero-frequency mode,

$$\lambda_1 = 0.$$

Therefore, solvability requires

$$\widehat{F}(1) = 0.$$

This is the discrete compatibility condition. The zero mode of the solution is not determined by the equation, so fixing

$$\widehat{U}(1)$$

restores uniqueness.

Remark 6.5.1. This is the discrete analogue of the pure Neumann problem: the constant mode lies in the null space, compatibility is required for the right-hand side, and an additional normalization determines the solution uniquely.

6.6 Stationary Iterative Methods

When a fast direct solver is unavailable, iterative methods can be used.

Consider

$$Ax = b.$$

A stationary iteration has the form

$$x^{(k+1)} = Tx^{(k)} + c.$$

Theorem 6.6.1 (Convergence of a Stationary Iteration)**.** *If*

$$\rho(T) < 1,$$

then

$$x^{(k)} \longrightarrow x$$

for every initial vector $x^{(0)}$, where x is the unique fixed point satisfying

$$x = Tx + c.$$

Define the error

$$e^{(k)} = x^{(k)} - x.$$

Then

$$e^{(k+1)} = Te^{(k)},$$

so

$$\boxed{e^{(k)} = T^k e^{(0)}}.$$

For any induced vector norm, if

$$\|T\| < 1,$$

then

$$\boxed{\|x - x^{(k)}\| \leq \|T\|^k \|x - x^{(0)}\|}.$$

An a posteriori estimate is

$$\boxed{\|x - x^{(k)}\| \leq \frac{\|T\|}{1 - \|T\|} \left\|x^{(k)} - x^{(k-1)}\right\|}.$$

If

$$T = R\Lambda R^{-1},$$

then

$$T^k = R\Lambda^k R^{-1}.$$

Hence,

$$\|e^{(k)}\|_2 \leq \|R\|_2 \|R^{-1}\|_2 \rho(T)^k \|e^{(0)}\|_2.$$

Define

$$\kappa_2(R) = \|R\|_2 \|R^{-1}\|_2.$$

Then

$$\boxed{\|e^{(k)}\|_2 \leq \kappa_2(R)\rho(T)^k \|e^{(0)}\|_2}.$$

Thus the asymptotic convergence rate is governed by the spectral radius $\rho(T)$.

For the five-point Poisson equation,

$$\frac{U_{i,j-1} + U_{i-1,j} - 4U_{ij} + U_{i+1,j} + U_{i,j+1}}{h^2} = f_{ij}.$$

Solving for U_{ij} gives

$$U_{ij} = \frac{1}{4}\left(U_{i,j-1} + U_{i-1,j} + U_{i+1,j} + U_{i,j+1} - h^2 f_{ij}\right).$$

The Jacobi method uses only values from iteration k:

$$U_{ij}^{(k+1)} = \frac{1}{4}\left(U_{i,j-1}^{(k)} + U_{i-1,j}^{(k)} + U_{i+1,j}^{(k)} + U_{i,j+1}^{(k)} - h^2 f_{ij}\right).$$

The Gauss–Seidel method uses newly computed values immediately:

$$U_{ij}^{(k+1)} = \frac{1}{4}\left(U_{i,j-1}^{(k+1)} + U_{i-1,j}^{(k+1)} + U_{i+1,j}^{(k)} + U_{i,j+1}^{(k)} - h^2 f_{ij}\right).$$

The successive over-relaxation method is

$$U^{(k+1)} = (1 - \omega)U^{(k)} + \omega U_{\mathrm{GS}}^{(k+1)},$$

where

$$1 < \omega < 2.$$

For the standard Poisson problem, the optimal relaxation parameter recorded in the notes is

$$\boxed{\omega_{\mathrm{opt}} = \frac{2}{1 + \sin(\pi h)}}.$$

For the one-dimensional Poisson problem, write

$$A = D - L - U.$$

The Jacobi iteration matrix is

$$T_J = D^{-1}(L+U) = I - D^{-1}A.$$

Since

$$D = -\frac{2}{h^2}I,$$

we obtain

$$T_J = I + \frac{h^2}{2}A.$$

Therefore,

$$\lambda_p(T_J) = 1 + \frac{h^2}{2}\lambda_p(A).$$

Using the eigenvalues of the one-dimensional discrete Laplacian,

$$\boxed{\lambda_p(T_J) = \cos(p\pi h)}.$$

Hence,

$$\rho(T_J) = \cos(\pi h).$$

For small h,

$$\rho(T_J) = 1 - \frac{\pi^2}{2}h^2 + O(h^4).$$

Thus,

$$\rho(T_J) \longrightarrow 1$$

as $h \to 0$, so Jacobi becomes increasingly slow as the mesh is refined.

The notes also record that for a range of higher-frequency modes,

$$|\lambda_p(T_J)|$$

is significantly smaller than one. These components decay much more rapidly than the low-frequency components. Thus Jacobi acts as a smoother: oscillatory errors decay rapidly, while smooth errors decay slowly.

To reduce the error by a fixed factor, we need approximately

$$\rho(T_J)^k \approx \varepsilon.$$

Taking logarithms gives

$$k \approx \frac{\log \varepsilon}{\log \rho(T_J)}.$$

Since

$$\log \rho(T_J) \approx -\frac{\pi^2}{2}h^2,$$

we obtain

$$k = O\left(h^{-2}\log\frac{1}{\varepsilon}\right).$$

For a one-dimensional problem, one sweep costs $O(N)$, so the total cost is approximately

$$O\left(N^3 \log\frac{1}{\varepsilon}\right).$$

For a two-dimensional problem with N^2 unknowns, one sweep costs $O(N^2)$, while the number of iterations grows like $O(N^2)$. Thus the total cost is approximately

$$O\left(N^4 \log\frac{1}{\varepsilon}\right).$$

Remark 6.6.2. Simple stationary iterations are therefore inefficient as complete solvers on very fine meshes. However, the rapid damping of high-frequency errors is an important idea behind multigrid methods.

Chapter 7

Time Discretization for Ordinary Differential Equations

7.1 Initial Value Problems

Consider the initial value problem

$$\begin{cases} u'(t) = f(t, u(t)), \\ u(a) = \alpha, \end{cases}$$

where

$$u : (a, b) \longrightarrow \mathbb{R}^d, \qquad f : (a, b) \times \mathbb{R}^d \longrightarrow \mathbb{R}^d.$$

When

$$d = 1,$$

the problem is scalar.

A particularly important case is the linear system

$$u'(t) = A(t)u(t) + g(t),$$

where

$$A(t) \in \mathbb{R}^{d \times d}, \qquad g(t) \in \mathbb{R}^d.$$

If

$$g \equiv 0,$$

then the equation is homogeneous.

If A is constant, Duhamel's principle gives

$$u(t) = e^{(t-t_0)A}u(t_0) + \int_{t_0}^{t} e^{(t-s)A}g(s)\, ds.$$

Suppose f is continuous on

$$D = \{(t, u) : t_0 \leq t \leq t_1, \quad |u - \alpha| \leq a\},$$

and Lipschitz continuous in u, so that there exists $L > 0$ such that

$$|f(t, u_1) - f(t, u_2)| \leq L|u_1 - u_2|$$

for all

$$(t, u_1), (t, u_2) \in D.$$

Define

$$S = \max_{(t,u)\in D} |f(t, u)|.$$

Then one may choose

$$T^* = \min\left\{t_1, t_0 + \frac{a}{S}\right\}.$$

Under these assumptions, the IVP has a unique local solution on the corresponding interval.

Remark 7.1.1. The quantity S is a bound on the largest rate at which the solution may move away from its initial value.

7.2 One-Step Time Discretization

Consider

$$\begin{cases} u'(t) = f(t,u), \quad 0 \le t \le T, \\ u(0) = \alpha. \end{cases}$$

Introduce the time grid

$$\tau = \frac{T}{N}, \qquad t_n = n\tau,$$

and let

$$U_n \approx u(t_n).$$

Integrating the ODE gives

$$u(t_{n+1}) - u(t_n) = \int_{t_n}^{t_{n+1}} f(t,u(t))\,dt.$$

Different numerical methods arise from different approximations of this integral.

For Forward Euler, approximate the integrand at the left endpoint:

$$f(t,u(t)) \approx f(t_n,u(t_n)).$$

Then

$$u(t_{n+1}) \approx u(t_n) + \tau f(t_n,u(t_n)),$$

so

$$\boxed{U_{n+1} = U_n + \tau f(t_n,U_n)}.$$

This method is explicit.

For Backward Euler, approximate the integrand at the right endpoint:

$$f(t,u(t)) \approx f(t_{n+1},u(t_{n+1})).$$

Then

$$u(t_{n+1}) \approx u(t_n) + \tau f(t_{n+1},u(t_{n+1})),$$

so

$$\boxed{U_{n+1} = U_n + \tau f(t_{n+1},U_{n+1})}.$$

This method is implicit.

For the trapezoidal rule,

$$\int_{t_n}^{t_{n+1}} f(t,u(t))\,dt$$

is approximated by

$$\frac{\tau}{2}\left[f(t_n,u(t_n)) + f(t_{n+1},u(t_{n+1}))\right].$$

Therefore,

$$\boxed{U_{n+1} = U_n + \frac{\tau}{2}\left[f(t_n,U_n) + f(t_{n+1},U_{n+1})\right]}.$$

This method is implicit.

The Improved Euler method first predicts

$$\widehat{U}_{n+1} = U_n + \tau f(t_n,U_n),$$

and then corrects using the average of two slopes:

$$\boxed{U_{n+1} = U_n + \frac{\tau}{2}\left[f(t_n, U_n) + f(t_{n+1}, \widehat{U}_{n+1})\right]}.$$

This method is explicit.

A general explicit s-stage Runge–Kutta method has the form

$$K_1 = f(t_n, U_n),$$

and for

$$i = 2, \ldots, s,$$

$$K_i = f\left(t_n + c_i\tau, U_n + \tau \sum_{j=1}^{i-1} a_{ij} K_j\right).$$

The update is

$$\boxed{U_{n+1} = U_n + \tau \sum_{i=1}^{s} b_i K_i}.$$

7.3 Error, Consistency, and Convergence

Define the global error by

$$e_n = u(t_n) - U_n.$$

For a one-step method written as

$$U_{n+1} = U_n + \tau\phi(t_n, U_n, \tau),$$

define the local truncation error by

$$\tau_{n+1} = u(t_{n+1}) - u(t_n) - \tau\phi(t_n, u(t_n), \tau).$$

For Forward Euler, Taylor expansion gives

$$u(t_{n+1}) = u(t_n) + \tau u'(t_n) + \frac{\tau^2}{2} u''(\xi_n),$$

where

$$\xi_n \in (t_n, t_{n+1}).$$

Therefore,

$$\boxed{\tau_{n+1}^{\mathrm{FE}} = \frac{\tau^2}{2} u''(\xi_n)}.$$

For the trapezoidal rule,

$$\boxed{\tau_{n+1}^{\mathrm{TR}} = -\frac{\tau^3}{12} u^{(3)}(\xi_n)}.$$

Definition 7.3.1 (Consistency)**.** The method is consistent if

$$\max_{0 \le n \le N-1} \left| \frac{\tau_{n+1}}{\tau} \right| \longrightarrow 0$$

as

$$\tau \longrightarrow 0.$$

Definition 7.3.2 (Convergence)**.** The method is convergent if

$$\max_{0\le n\le N} |e_n| \longrightarrow 0$$

as

$$\tau \longrightarrow 0.$$

Definition 7.3.3 (Order of Accuracy)**.** The method has order p if

$$|e_n| = O(\tau^p).$$

Stability concerns perturbations of the initial value. Suppose

$$U_0 = \alpha, \qquad \widetilde{U}_0 = \alpha + \delta,$$

and

$$U_{n+1} = U_n + \tau\phi(t_n, U_n, \tau),$$
$$\widetilde{U}_{n+1} = \widetilde{U}_n + \tau\phi(t_n, \widetilde{U}_n, \tau).$$

Define

$$w_n = \widetilde{U}_n - U_n.$$

Then

$$w_{n+1} = w_n + \tau\left[\phi(t_n, \widetilde{U}_n, \tau) - \phi(t_n, U_n, \tau)\right].$$

If ϕ is Lipschitz continuous in u with constant L, then

$$|w_{n+1}| \le (1 + \tau L)|w_n|.$$

Therefore,

$$|w_n| \le (1 + \tau L)^n|\delta| \le e^{n\tau L}|\delta|.$$

Since

$$n\tau \le T,$$

we obtain

$$\boxed{|w_n| \le e^{LT}|\delta|}.$$

Thus the numerical solution is stable with respect to perturbations of the initial value.

Theorem 7.3.4. *Consider*

$$U_{n+1} = U_n + \tau\phi(t_n, U_n, \tau),$$

where ϕ is continuous and Lipschitz continuous in u. Then the method is convergent if and only if it is consistent.

For consistency,

$$\frac{\tau_{n+1}}{\tau} = \frac{u(t_{n+1}) - u(t_n)}{\tau} - \phi(t_n, u(t_n), \tau).$$

As

$$\tau \to 0,$$

the first term tends to

$$u'(t_n) = f(t_n, u(t_n)).$$

Therefore,

$$\frac{\tau_{n+1}}{\tau} \to 0$$

if and only if

$$\boxed{\phi(t,u,0) = f(t,u)}.$$

Suppose now that

$$|\tau_{n+1}| \le C\tau^{p+1}.$$

The exact and numerical solutions satisfy

$$u(t_{n+1}) = u(t_n) + \tau\phi(t_n, u(t_n), \tau) + \tau_{n+1},$$

and

$$U_{n+1} = U_n + \tau\phi(t_n, U_n, \tau).$$

Subtracting gives

$$e_{n+1} = e_n + \tau\left[\phi(t_n, u(t_n), \tau) - \phi(t_n, U_n, \tau)\right] + \tau_{n+1}.$$

Hence,

$$|e_{n+1}| \le (1+\tau L)|e_n| + C\tau^{p+1}.$$

Iterating and assuming the initial value is exact,

$$e_0 = 0,$$

gives

$$|e_n| \le C\tau^{p+1}\sum_{j=0}^{n-1}(1+\tau L)^j.$$

Using the geometric sum,

$$|e_n| \le \frac{C}{L}\left[(1+\tau L)^n - 1\right]\tau^p.$$

Since

$$(1+\tau L)^n \le e^{LT},$$

we obtain

$$\boxed{|e_n| \le \frac{C}{L}\left(e^{LT} - 1\right)\tau^p}.$$

Thus local truncation error $O(\tau^{p+1})$ leads to global error $O(\tau^p)$.

7.4 Multistep Methods and ODE Systems

A general linear multistep method has the form

$$y_{n+1} = a_{m-1}y_n + a_{m-2}y_{n-1} + \cdots + a_0 y_{n+1-m}$$

$$+\tau\left[b_m f(t_{n+1}, y_{n+1}) + \cdots + b_0 f(t_{n+1-m}, y_{n+1-m})\right].$$

If

$$b_m = 0,$$

the method is explicit. If

$$b_m \neq 0,$$

the method is implicit.

The notes also record the approximation

$$2f(t_n, U_n) - f(t_{n-1}, U_{n-1}),$$

which uses information from two previous time levels and is a typical ingredient in multistep constructions.

Higher-order ODEs can be rewritten as first-order systems. Consider

$$y^{(m)} = f\left(t, y, y', \ldots, y^{(m-1)}\right),$$

with initial conditions

$$y(0) = a_0, \qquad y'(0) = a_1, \qquad \ldots, \qquad y^{(m-1)}(0) = a_{m-1}.$$

Define

$$u_1 = y, \qquad u_2 = y', \qquad \ldots, \qquad u_m = y^{(m-1)}.$$

Then

$$u_1' = u_2, \qquad u_2' = u_3, \qquad \ldots, \qquad u_{m-1}' = u_m,$$

and

$$u_m' = f(t, u_1, \ldots, u_m).$$

Thus,

$$\boxed{U' = F(t, U)}.$$

Any one-step method can therefore be applied directly to the resulting first-order system.

7.5 Strong Stability and the Test Equation

A numerical method is strongly stable for a dissipative problem if

$$\boxed{|U_{n+1}| \le |U_n|}.$$

Consider the test equation

$$u' = \lambda u, \qquad u(0) = \alpha, \qquad \operatorname{Re}\lambda < 0.$$

The exact solution is

$$u(t) = \alpha e^{\lambda t},$$

so

$$|u(t)|$$

decreases in time.

For Forward Euler,

$$U_{n+1} = U_n + \tau\lambda U_n.$$

Define

$$z = \tau\lambda.$$

Then

$$U_{n+1} = (1 + z)U_n.$$

Strong stability requires

$$\boxed{|1+z| \leq 1}.$$

For Backward Euler,

$$U_{n+1} = U_n + \tau\lambda U_{n+1}.$$

Therefore,

$$U_{n+1} = \frac{1}{1-z} U_n.$$

Strong stability requires

$$\boxed{|1-z| \geq 1}.$$

For the trapezoidal rule,

$$U_{n+1} = U_n + \frac{z}{2}\left(U_n + U_{n+1}\right).$$

Hence,

$$U_{n+1} = \frac{1+\frac{z}{2}}{1-\frac{z}{2}} U_n.$$

Strong stability requires

$$\boxed{\left|\frac{1+\frac{z}{2}}{1-\frac{z}{2}}\right| \leq 1}.$$

The corresponding stability functions are

$$R_{\mathrm{FE}}(z) = 1 + z,$$

$$R_{\mathrm{BE}}(z) = \frac{1}{1-z},$$

and

$$R_{\mathrm{TR}}(z) = \frac{1+\frac{z}{2}}{1-\frac{z}{2}}.$$

Definition 7.5.1 (Absolute Stability Region)**.** The absolute stability region is

$$\mathcal{S} = \{z \in \mathbb{C} : |R(z)| \leq 1\}.$$

Forward Euler has a bounded stability region. Backward Euler and the trapezoidal rule contain the entire left half-plane.

Consider the example

$$u' = -100u, \qquad u(0) = 1.$$

The exact solution is

$$u(t) = e^{-100t}.$$

Forward Euler gives

$$U_{n+1} = (1 - 100\tau)U_n.$$

Strong stability requires

$$|1 - 100\tau| \leq 1,$$

so

$$\boxed{0 \leq \tau \leq \frac{1}{50}}.$$

Backward Euler gives

$$U_{n+1} = \frac{1}{1+100\tau} U_n,$$

which is stable for every

$$\tau > 0.$$

The trapezoidal rule gives

$$U_{n+1} = \frac{1-50\tau}{1+50\tau} U_n,$$

which is also stable for every

$$\tau > 0.$$

7.6 Stability of Linear Systems

Consider

$$U'(t) = AU(t), \qquad U(0) = \alpha.$$

Suppose

$$A = X^{-1}\Lambda X,$$

where

$$\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_d),$$

and

$$\operatorname{Re}\lambda_j < 0.$$

Define

$$V = XU.$$

Then

$$V' = \Lambda V,$$

so each component satisfies

$$V_j' = \lambda_j V_j.$$

Applying Forward Euler gives

$$U^{n+1} = U^n + \tau A U^n.$$

Multiplying by X,

$$V^{n+1} = (I + \tau\Lambda)V^n.$$

Therefore,

$$V_j^{n+1} = (1 + \tau\lambda_j)V_j^n.$$

Stability requires

$$|1 + \tau\lambda_j| \le 1$$

for every eigenvalue.

For real negative eigenvalues,

$$-2 \le \tau\lambda_j \le 0.$$

Hence,

$$\boxed{\tau \le \min_j \frac{-2}{\lambda_j}}.$$

Thus the most negative eigenvalue determines the maximum stable time step for Forward Euler.

Remark 7.6.1. This observation becomes essential for parabolic PDEs. After spatial discretization, the most negative eigenvalue of the discrete Laplacian typically has magnitude $O(h^{-2})$, so an explicit method inherits a time-step restriction of order

$$\tau = O(h^2).$$

This is the starting point of Chapter 8.

Chapter 8

Numerical Methods for Parabolic Equations

8.1 Heat Equation and Spatial Semidiscretization

Consider the one-dimensional heat equation

$$\begin{cases} u_t = au_{xx} + f(x,t), & t \in [0,T], \quad x \in (0,1), \\ u(0,t) = u(1,t) = 0, \\ u(x,0) = u_0(x). \end{cases}$$

Introduce the uniform spatial grid

$$h = \frac{1}{N+1}, \qquad x_i = ih, \qquad i = 0, 1, \ldots, N+1.$$

Let

$$U_i(t) \approx u(x_i, t), \qquad f_i(t) = f(x_i, t).$$

The boundary conditions give

$$U_0(t) = U_{N+1}(t) = 0.$$

Using the centered difference approximation,

$$u_{xx}(x_i, t) \approx \frac{U_{i-1}(t) - 2U_i(t) + U_{i+1}(t)}{h^2},$$

we obtain

$$\frac{dU_i}{dt} = a\frac{U_{i-1} - 2U_i + U_{i+1}}{h^2} + f_i(t).$$

Define

$$U(t) = \begin{bmatrix} U_1(t) \\ U_2(t) \\ \vdots \\ U_N(t) \end{bmatrix}, \qquad F(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_N(t) \end{bmatrix}.$$

Then

$$\boxed{U'(t) = AU(t) + F(t)},$$

where

$$A = \frac{a}{h^2}\begin{bmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -2 \end{bmatrix}.$$

Thus spatial discretization transforms the PDE into a system of ordinary differential equations. This procedure is usually called the *method of lines*.

8.2 Time Discretization and Truncation Error

Introduce the time grid

$$\tau = \frac{T}{M}, \qquad t_n = n\tau,$$

and let

$$U_i^n \approx U_i(t_n) \approx u(x_i, t_n).$$

Forward Euler gives

$$\frac{U_i^{n+1} - U_i^n}{\tau} = a\frac{U_{i-1}^n - 2U_i^n + U_{i+1}^n}{h^2} + f_i^n.$$

In matrix form,

$$\boxed{U^{n+1} = (I + \tau A)U^n + \tau F^n}.$$

Backward Euler gives

$$\frac{U_i^{n+1} - U_i^n}{\tau} = a\frac{U_{i-1}^{n+1} - 2U_i^{n+1} + U_{i+1}^{n+1}}{h^2} + f_i^{n+1}.$$

Therefore,

$$(I - \tau A)U^{n+1} = U^n + \tau F^{n+1},$$

so

$$\boxed{U^{n+1} = (I - \tau A)^{-1}\left(U^n + \tau F^{n+1}\right)}.$$

Crank–Nicolson gives

$$\frac{U_i^{n+1} - U_i^n}{\tau} = \frac{a}{2}\left(\delta_h^2 U_i^n + \delta_h^2 U_i^{n+1}\right) + \frac{1}{2}\left(f_i^n + f_i^{n+1}\right),$$

where

$$\delta_h^2 U_i = \frac{U_{i-1} - 2U_i + U_{i+1}}{h^2}.$$

In matrix form,

$$\frac{U^{n+1} - U^n}{\tau} = \frac{1}{2}A\left(U^n + U^{n+1}\right) + \frac{1}{2}\left(F^n + F^{n+1}\right).$$

Hence,

$$\left(I - \frac{\tau}{2}A\right)U^{n+1} = \left(I + \frac{\tau}{2}A\right)U^n + \frac{\tau}{2}\left(F^n + F^{n+1}\right),$$

and therefore

$$\boxed{U^{n+1} = \left(I - \frac{\tau}{2}A\right)^{-1}\left[\left(I + \frac{\tau}{2}A\right)U^n + \frac{\tau}{2}\left(F^n + F^{n+1}\right)\right]}.$$

We now estimate the local truncation error. For Forward Euler, substitute the exact solution into the discrete equation:

$$\xi_i^{n+1} = u(x_i, t_{n+1}) - u(x_i, t_n)$$
$$-\tau\left[a\frac{u(x_{i-1}, t_n) - 2u(x_i, t_n) + u(x_{i+1}, t_n)}{h^2} + f(x_i, t_n)\right].$$

Taylor expansion in time gives

$$u(x_i, t_{n+1}) = u(x_i, t_n) + \tau u_t(x_i, t_n) + \frac{\tau^2}{2}u_{tt}(x_i, t_n) + O(\tau^3).$$

Taylor expansion in space gives

$$\frac{u(x_{i-1}, t_n) - 2u(x_i, t_n) + u(x_{i+1}, t_n)}{h^2} = u_{xx}(x_i, t_n) + \frac{h^2}{12}u_{xxxx}(x_i, t_n) + O(h^4).$$

Since

$$u_t - au_{xx} - f = 0,$$

we obtain

$$\xi_i^{n+1} = \tau \left[\frac{\tau}{2} u_{tt} - \frac{ah^2}{12} u_{xxxx} + O(\tau^2) + O(h^4) \right].$$

Therefore,

$$\boxed{\xi_i^{n+1} = O\left(\tau(\tau + h^2)\right)}.$$

If the solution is sufficiently smooth, the notes record

$$|\xi^{n+1}| \leq C\tau(\tau + h^2)$$

for Forward Euler and Backward Euler, while

$$|\xi^{n+1}| \leq C\tau(\tau^2 + h^2)$$

for Crank–Nicolson.

Thus the expected global orders are

$$\begin{aligned} \text{Forward Euler} &: O(\tau + h^2), \\ \text{Backward Euler} &: O(\tau + h^2), \\ \text{Crank–Nicolson} &: O(\tau^2 + h^2). \end{aligned}$$

8.3 Consistency, Stability, and Convergence

Let

$$\widehat{U}^n = \begin{bmatrix} u(x_1, t_n) \\ u(x_2, t_n) \\ \vdots \\ u(x_N, t_n) \end{bmatrix}$$

denote the exact solution sampled at the grid points. Define the global error by

$$E^n = \widehat{U}^n - U^n.$$

Definition 8.3.1 (Consistency)**.** The finite difference method is consistent if

$$\frac{\|\xi^{n+1}\|}{\tau} \longrightarrow 0$$

as

$$\tau, h \longrightarrow 0.$$

Definition 8.3.2 (Convergence)**.** The method is convergent if

$$\|E^n\| \longrightarrow 0$$

as

$$\tau, h \longrightarrow 0.$$

A linear time-stepping method can be written as

$$U^{n+1} = BU^n + b^n.$$

The exact grid values satisfy

$$\widehat{U}^{n+1} = B\widehat{U}^n + b^n + D\xi^{n+1}.$$

Subtracting gives

$$E^{n+1} = BE^n + D\xi^{n+1}.$$

Iterating,

$$\boxed{E^n = B^n E^0 + \sum_{k=1}^{n} B^{n-k} D\xi^k}.$$

Definition 8.3.3 (Lax–Richtmyer Stability)**.** The method

$$U^{n+1} = BU^n + b^n$$

is Lax–Richtmyer stable if there exists a constant $C > 0$, independent of h, τ, and n, such that

$$\boxed{\|B^n\| \le C}$$

whenever

$$n\tau \le T.$$

Indeed,

$$U^n = B^n U^0 + \sum_{k=0}^{n-1} B^k b^{n-1-k},$$

so stability gives

$$\|U^n\| \le C\|U^0\| + C\sum_{k=0}^{n-1} \|b^k\|.$$

To interpret stability as sensitivity to perturbations, suppose

$$\widetilde{U}^0 = U^0 + \delta$$

and

$$\widetilde{U}^{n+1} = B\widetilde{U}^n + b^n.$$

Define

$$W^n = \widetilde{U}^n - U^n.$$

Then

$$W^{n+1} = BW^n,$$

so

$$W^n = B^n \delta.$$

Therefore,

$$\boxed{\|W^n\| \le C\|\delta\|}.$$

A sufficient condition is

$$\|B\| \le 1 + \alpha\tau.$$

Then

$$\|B^n\| \le (1 + \alpha\tau)^n \le e^{\alpha T}.$$

Theorem 8.3.4 (Lax Equivalence Theorem)**.** *For a well-posed linear initial value problem, a consistent linear finite difference method is convergent if and only if it is Lax–Richtmyer stable.*

The error formula

$$E^n = B^n E^0 + \sum_{k=1}^{n} B^{n-k} D\xi^k$$

shows the basic mechanism. Stability uniformly controls the powers of B, while consistency forces the truncation error to zero.

A stronger condition is

$$\boxed{\|B\| \leq 1}.$$

Definition 8.3.5 (Strong Stability)**.** A linear time-stepping method is strongly stable in a given norm if

$$\|B\| \leq 1.$$

Strong stability immediately implies Lax–Richtmyer stability because

$$\|B^n\| \leq \|B\|^n \leq 1.$$

8.4 Norm Stability and von Neumann Analysis

For the one-dimensional heat equation, the eigenvalues of the spatial matrix are

$$\lambda_p = \frac{2a}{h^2}\left[\cos(p\pi h) - 1\right], \qquad p = 1, \ldots, N.$$

Thus

$$\lambda_p < 0.$$

For Forward Euler,

$$B_{\mathrm{FE}} = I + \tau A.$$

Since A is symmetric,

$$\|B_{\mathrm{FE}}\|_2 = \max_p |1 + \tau\lambda_p|.$$

Strong stability requires

$$-2 \leq \tau\lambda_p \leq 0$$

for every p. Since the most negative eigenvalue has magnitude approximately $4a/h^2$,

$$\boxed{\frac{a\tau}{h^2} \leq \frac{1}{2}}.$$

For Backward Euler,

$$B_{\mathrm{BE}} = (I - \tau A)^{-1}.$$

Its eigenvalues are

$$\frac{1}{1 - \tau\lambda_p}.$$

Since

$$\lambda_p < 0,$$

we have

$$\left|\frac{1}{1-\tau\lambda_p}\right| \le 1.$$

Therefore,

$$\boxed{\|B_{\mathrm{BE}}\|_2 \le 1}$$

for every $\tau > 0$.

For Crank–Nicolson,

$$B_{\mathrm{CN}} = \left(I - \frac{\tau}{2}A\right)^{-1}\left(I + \frac{\tau}{2}A\right).$$

Its eigenvalues are

$$\frac{1+\frac{\tau}{2}\lambda_p}{1-\frac{\tau}{2}\lambda_p}.$$

Since $\lambda_p < 0$,

$$\left|\frac{1+\frac{\tau}{2}\lambda_p}{1-\frac{\tau}{2}\lambda_p}\right| \le 1.$$

Hence,

$$\boxed{\|B_{\mathrm{CN}}\|_2 \le 1}.$$

Thus Backward Euler and Crank–Nicolson are unconditionally strongly stable in the discrete L^2-norm, while Forward Euler requires

$$\tau = O(h^2).$$

We now consider the discrete L^∞-norm. For Forward Euler, define

$$r = \frac{a\tau}{h^2}.$$

Then

$$U_i^{n+1} = rU_{i-1}^n + (1-2r)U_i^n + rU_{i+1}^n + \tau f_i^n.$$

If

$$0 \le r \le \frac{1}{2},$$

the coefficients in the homogeneous part are nonnegative and sum to one. Therefore,

$$\boxed{\|B_{\mathrm{FE}}\|_\infty \le 1}.$$

For Backward Euler,

$$B_{\mathrm{BE}} = (I - \tau A)^{-1}.$$

The notes use the discrete maximum principle to show

$$(I - \tau A)^{-1} \ge 0.$$

Let

$$S = (I - \tau A)^{-1}\mathbf{1}.$$

Then

$$(I - \tau A)S = \mathbf{1}.$$

Using the comparison principle gives

$$0 \le S \le \mathbf{1}.$$

Hence,

$$\boxed{\|B_{\mathrm{BE}}\|_\infty \le 1}.$$

For Crank–Nicolson, the L^2-stability is unconditional. The simple L^∞ strong-stability argument in the notes requires the additional restriction

$$\boxed{\tau \le h^2}$$

for the normalized equation with $a = 1$.

Von Neumann analysis gives another approach to stability. Consider the Fourier mode

$$W_j = e^{ijh\xi}.$$

For the centered first difference,

$$D_0 W_j = \frac{W_{j+1} - W_{j-1}}{2h} = i\frac{\sin(h\xi)}{h} W_j.$$

For the centered second difference,

$$D_0^2 W_j = \frac{W_{j-1} - 2W_j + W_{j+1}}{h^2} = \frac{2\cos(h\xi) - 2}{h^2} W_j.$$

A discrete grid function has the Fourier representation

$$U_j = \frac{1}{\sqrt{2\pi}} \int_{-\pi/h}^{\pi/h} \widehat{U}(\xi) e^{ijh\xi}\, d\xi,$$

with inverse transform

$$\widehat{U}(\xi) = \frac{h}{\sqrt{2\pi}} \sum_{j=-\infty}^{\infty} U_j e^{-ijh\xi}.$$

Parseval's relation gives

$$\boxed{\|U\|_2 = \|\widehat{U}\|_2}.$$

Suppose one time step maps a Fourier mode according to

$$e^{ijh\xi} \longmapsto g(\xi) e^{ijh\xi}.$$

The function

$$g(\xi)$$

is called the *amplification factor*. Strong L^2-stability follows if

$$|g(\xi)| \le 1$$

for every ξ. More generally, Lax–Richtmyer stability follows from

$$|g(\xi)| \le 1 + \alpha\tau,$$

because

$$|g(\xi)|^n \le (1 + \alpha\tau)^n \le e^{\alpha T}.$$

Combining stability with the truncation error estimates gives

$$\boxed{\|E^n\| \le C(\tau^p + h^2)},$$

where

$$p = 1$$

for Forward Euler and Backward Euler, while

$$p = 2$$

for Crank–Nicolson. :contentReference[oaicite:4]index=4

8.5 Multidimensional Parabolic Equations

Consider

$$\begin{cases} u_t = \Delta u + f, & t \in (0,T], \quad (x,y) \in \Omega, \\ u(x,y,0) = u_0(x,y), & (x,y) \in \Omega, \\ u(x,y,t) = g(x,y,t), & (x,y) \in \partial\Omega. \end{cases}$$

For simplicity, take

$$g \equiv 0, \qquad \Omega = [0,1]^2.$$

Introduce the uniform grid

$$h = \frac{1}{N+1}, \qquad x_i = ih, \qquad y_j = jh,$$

and let

$$U_{ij}(t) \approx u(x_i, y_j, t).$$

The discrete Laplacian is

$$\Delta_h U_{ij} = \frac{U_{i-1,j} + U_{i+1,j} + U_{i,j-1} + U_{i,j+1} - 4U_{ij}}{h^2}.$$

Let D denote the one-dimensional second-difference matrix. Then the two-dimensional discrete operator is

$$\boxed{D_2 = I_N \otimes D + D \otimes I_N}.$$

Thus the semidiscrete problem is

$$U' = D_2 U + F.$$

Forward Euler gives

$$U^{n+1} = (I + \tau D_2)U^n + \tau F^n.$$

Backward Euler gives

$$U^{n+1} = (I - \tau D_2)^{-1}\left(U^n + \tau F^{n+1}\right).$$

Crank–Nicolson gives

$$U^{n+1} = \left(I - \frac{\tau}{2}D_2\right)^{-1}\left[\left(I + \frac{\tau}{2}D_2\right)U^n + \frac{\tau}{2}\left(F^n + F^{n+1}\right)\right].$$

The eigenvalues of D_2 are

$$\lambda_{p,q} = \frac{2}{h^2}\left[\cos(p\pi h) + \cos(q\pi h) - 2\right].$$

Backward Euler and Crank–Nicolson are again unconditionally stable in the discrete L^2-norm.

For Forward Euler,

$$|1 + \tau\lambda_{p,q}| \leq 1$$

must hold for every p, q. Since the most negative eigenvalue is approximately

$$-\frac{8}{h^2},$$

we obtain

$$\boxed{\tau \leq \frac{h^2}{4}}.$$

The notes therefore record the same global error structure:

$$O\left(\tau^p + h^2\right),$$

with $p = 1$ for FE and BE and $p = 2$ for CN. :contentReference[oaicite:5]index=5

8.6 Solvers for Multidimensional Implicit Systems

The two-dimensional Dirichlet Laplacian can be diagonalized using a two-dimensional discrete sine transform.

For a grid function $X(m, n)$, define

$$Y(p, q) = \sum_{m=1}^{N} \sum_{n=1}^{N} X(m, n) \sin\left(\frac{pm\pi}{N+1}\right) \sin\left(\frac{qn\pi}{N+1}\right).$$

The inverse transform is

$$X(m, n) = \frac{4}{(N+1)^2} \sum_{p=1}^{N} \sum_{q=1}^{N} Y(p, q) \sin\left(\frac{pm\pi}{N+1}\right) \sin\left(\frac{qn\pi}{N+1}\right).$$

Using fast transforms, the cost is approximately

$$O(N^2 \log N).$$

For example, consider the Backward Euler system

$$(I - \tau D_2)U^{n+1} = G.$$

Expand

$$U_{ij}^{n+1} = \sum_{p=1}^{N} \sum_{q=1}^{N} \widehat{U}_{p,q}^{n+1} \sin\left(\frac{pi\pi}{N+1}\right) \sin\left(\frac{qj\pi}{N+1}\right),$$

and similarly

$$G_{ij} = \sum_{p=1}^{N} \sum_{q=1}^{N} \widehat{G}_{p,q} \sin\left(\frac{pi\pi}{N+1}\right) \sin\left(\frac{qj\pi}{N+1}\right).$$

Then

$$(1 - \tau\lambda_{p,q})\, \widehat{U}_{p,q}^{n+1} = \widehat{G}_{p,q},$$

so

$$\boxed{\widehat{U}_{p,q}^{n+1} = \frac{\widehat{G}_{p,q}}{1 - \tau\lambda_{p,q}}}.$$

Simple iterative methods may also be used. Consider

$$U - \tau D_2 U = G$$

and define

$$\sigma = \frac{\tau}{h^2}.$$

At an interior point,

$$(1 + 4\sigma)U_{ij} - \sigma\left(U_{i-1,j} + U_{i+1,j} + U_{i,j-1} + U_{i,j+1}\right) = G_{ij}.$$

The Jacobi iteration is

$$U_{ij}^{(k+1)} = \frac{1}{1 + 4\sigma}\left[\sigma\left(U_{i-1,j}^{(k)} + U_{i+1,j}^{(k)} + U_{i,j-1}^{(k)} + U_{i,j+1}^{(k)}\right) + G_{ij}\right].$$

The Gauss–Seidel method replaces already updated neighboring values by their newest available iterates.

The condition number of the implicit matrix is

$$\kappa_2(I - \tau D_2) = \frac{\max_{p,q} |1 - \tau\lambda_{p,q}|}{\min_{p,q} |1 - \tau\lambda_{p,q}|}.$$

Since

$$|\lambda_{\max}(D_2)| = O(h^{-2}),$$

the notes obtain the behavior

$$\boxed{\kappa_2(I - \tau D_2) = O\left(1 + \frac{\tau}{h^2}\right)}.$$

Thus a large ratio τ/h^2 can make the implicit linear system more difficult to solve iteratively. :contentReference[oaicite:6]index=6

8.7 Operator Splitting Methods

Consider

$$U' = (A + B)U.$$

Split the original problem into

$$V' = AV$$

and

$$W' = BW.$$

Starting from

$$V^n = U^n,$$

solve the first problem for one time step:

$$V^{n+1} = e^{\tau A} U^n.$$

Use this result as the initial value for the second problem:

$$U^{n+1} = e^{\tau B} V^{n+1}.$$

Therefore,

$$U^{n+1} = e^{\tau B} e^{\tau A} U^n.$$

In general,

$$e^{\tau B} e^{\tau A} \neq e^{\tau (A+B)}$$

because A and B need not commute.

Define the splitting error by

$$SE = e^{\tau (A+B)} - e^{\tau B} e^{\tau A}.$$

Taylor expansion gives

$$e^{\tau (A+B)} = I + \tau (A+B) + \frac{\tau^2}{2} (A+B)^2 + O(\tau^3),$$

while

$$e^{\tau B} e^{\tau A} = I + \tau (A+B) + \tau^2 \left(\frac{1}{2} A^2 + BA + \frac{1}{2} B^2 \right) + O(\tau^3).$$

Therefore,

$$\boxed{SE = \frac{\tau^2}{2} (AB - BA) + O(\tau^3)}.$$

Thus the leading splitting error is determined by the commutator

$$AB - BA.$$

A symmetric modification is

$$\boxed{U^{n+1} = e^{\frac{\tau}{2} A} e^{\tau B} e^{\frac{\tau}{2} A} U^n}.$$

This is the *Strang splitting method*. Its local splitting error is

$$O(\tau^3).$$

For the two-dimensional heat equation, write

$$U' = D_x U + D_y U,$$

where

$$D_x = I_N \otimes D, \qquad D_y = D \otimes I_N.$$

The direct Crank–Nicolson discretization is

$$\left[I - \frac{\tau}{2} (D_x + D_y) \right] U^{n+1} = \left[I + \frac{\tau}{2} (D_x + D_y) \right] U^n.$$

The matrix contains both spatial directions simultaneously, so splitting methods replace this large solve by successive one-dimensional solves.

For the alternating direction implicit method, introduce U^*:

$$U^* = U^n + \frac{\tau}{2} \left(D_x U^* + D_y U^n \right),$$

$$U^{n+1} = U^* + \frac{\tau}{2}\left(D_x U^* + D_y U^{n+1}\right).$$

Therefore,

$$\left(I - \frac{\tau}{2}D_x\right)U^* = \left(I + \frac{\tau}{2}D_y\right)U^n,$$

and

$$\left(I - \frac{\tau}{2}D_y\right)U^{n+1} = \left(I + \frac{\tau}{2}D_x\right)U^*.$$

Hence,

$$\boxed{U^{n+1} = \left(I - \frac{\tau}{2}D_y\right)^{-1}\left(I + \frac{\tau}{2}D_x\right)\left(I - \frac{\tau}{2}D_x\right)^{-1}\left(I + \frac{\tau}{2}D_y\right)U^n}.$$

The locally one-dimensional method uses

$$U^* = U^n + \frac{\tau}{2}D_x\left(U^n + U^*\right),$$

followed by

$$U^{n+1} = U^* + \frac{\tau}{2}D_y\left(U^* + U^{n+1}\right).$$

Therefore,

$$\boxed{U^{n+1} = \left(I - \frac{\tau}{2}D_y\right)^{-1}\left(I + \frac{\tau}{2}D_y\right)\left(I - \frac{\tau}{2}D_x\right)^{-1}\left(I + \frac{\tau}{2}D_x\right)U^n}.$$

The notes finally give a symmetric half-step/full-step/half-step splitting. The handwritten title "Strong Splitting" is corrected here to the standard name *Strang splitting*.

First solve in the x-direction for a half time step:

$$V^0 = U^n,$$

$$V^1 = V^0 + \frac{\tau}{4}D_x\left(V^0 + V^1\right).$$

Then solve in the y-direction for a full time step:

$$W^0 = V^1,$$

$$W^1 = W^0 + \frac{\tau}{2}D_y\left(W^0 + W^1\right).$$

Finally, solve again in the x-direction for a half time step:

$$\widetilde{V}^0 = W^1,$$

$$\widetilde{V}^1 = \widetilde{V}^0 + \frac{\tau}{4}D_x\left(\widetilde{V}^0 + \widetilde{V}^1\right).$$

Set

$$U^{n+1} = \widetilde{V}^1.$$

Therefore,

$$U^{n+1} = \left(I - \frac{\tau}{4}D_x\right)^{-1}\left(I + \frac{\tau}{4}D_x\right)$$
$$\times \left(I - \frac{\tau}{2}D_y\right)^{-1}\left(I + \frac{\tau}{2}D_y\right)\left(I - \frac{\tau}{4}D_x\right)^{-1}\left(I + \frac{\tau}{4}D_x\right)U^n.$$

The local truncation error is

$$O(\tau^3).$$

Remark 8.7.1. For ADI, the intermediate value U^* has a natural interpretation as an approximation at the intermediate stage. For LOD and Strang splitting, the intermediate variables are mainly computational quantities, so boundary conditions must be handled carefully. The notes emphasize that proper boundary treatment can preserve second-order accuracy.

Chapter 9

Galerkin and Finite Element Methods for Parabolic Equations

9.1 Weak Formulation and Galerkin Semidiscretization

Define the L^2-inner product by

$$(f, g) = \int_\Omega f(x)g(x)\,dx.$$

For homogeneous Dirichlet boundary conditions, define

$$H_0^1(\Omega) = \left\{u \in H^1(\Omega) : u|_{\partial\Omega} = 0\right\},$$

with norm

$$\|u\|_{H^1(\Omega)} = \left(\int_\Omega |u|^2 + |\nabla u|^2\,dx\right)^{1/2}.$$

Consider the parabolic problem

$$\begin{cases} u_t = \Delta u + f, & t \in (0, T], \quad x \in \Omega, \\ u(x, 0) = u_0(x), & x \in \Omega, \\ u(x, t) = 0, & x \in \partial\Omega. \end{cases}$$

Multiply the equation by

$$v \in H_0^1(\Omega)$$

and integrate over Ω:

$$(u_t, v) = (\Delta u, v) + (f, v).$$

Integration by parts gives

$$(\Delta u, v) = -(\nabla u, \nabla v),$$

because $v = 0$ on $\partial\Omega$. Therefore, the weak formulation is to find

$$u(t) \in H_0^1(\Omega)$$

such that

$$\boxed{(u_t, v) + (\nabla u, \nabla v) = (f, v)}$$

for every

$$v \in H_0^1(\Omega),$$

with

$$u(x, 0) = u_0(x).$$

Choose a finite-dimensional space

$$V_N \subset H_0^1(\Omega).$$

The Galerkin approximation seeks

$$u_N(t) \in V_N$$

such that

$$(\partial_t u_N, v_N) + (\nabla u_N, \nabla v_N) = (f, v_N)$$

for every

$$v_N \in V_N.$$

The initial condition is chosen as

$$u_N(x,0) = I_N u_0(x),$$

where

$$I_N : H_0^1(\Omega) \longrightarrow V_N$$

is the projection used in the notes.

It is characterized by

$$(u_0 - I_N u_0, v_N) = 0$$

for every

$$v_N \in V_N.$$

Equivalently, $I_N u_0$ minimizes the L^2-error over V_N.

9.2 Mass and Stiffness Matrices

Let

$$V_N = \text{span}\left\{\phi_1, \phi_2, \ldots, \phi_N\right\}.$$

Write

$$u_N(x,t) = \sum_{j=1}^{N} c_j(t)\phi_j(x).$$

Then

$$\partial_t u_N(x,t) = \sum_{j=1}^{N} c_j'(t)\phi_j(x).$$

Taking

$$v_N = \phi_i$$

in the weak formulation gives

$$(\partial_t u_N, \phi_i) + (\nabla u_N, \nabla \phi_i) = (f, \phi_i).$$

Substituting the basis expansion gives

$$\sum_{j=1}^{N} c_j'(t)(\phi_j, \phi_i) + \sum_{j=1}^{N} c_j(t)(\nabla\phi_j, \nabla\phi_i) = (f, \phi_i).$$

The initial condition gives

$$\sum_{j=1}^{N} c_j(0)(\phi_j, \phi_i) = (u_0, \phi_i).$$

Define the mass matrix by

$$\boxed{M_{ij} = (\phi_j, \phi_i)}$$

and the stiffness matrix by

$$\boxed{A_{ij} = (\nabla\phi_j, \nabla\phi_i)}.$$

Define

$$C(t) = \begin{bmatrix} c_1(t) \\ c_2(t) \\ \vdots \\ c_N(t) \end{bmatrix}, \qquad F(t) = \begin{bmatrix} (f, \phi_1) \\ (f, \phi_2) \\ \vdots \\ (f, \phi_N) \end{bmatrix}.$$

Then the semidiscrete Galerkin problem becomes

$$\boxed{MC'(t) + AC(t) = F(t)}.$$

The initial coefficients satisfy

$$MC(0) = C^0,$$

where

$$C_i^0 = (u_0, \phi_i).$$

Both M and A are symmetric. For any nonzero vector z,

$$z^T M z = \left\| \sum_j z_j \phi_j \right\|_{L^2}^2 > 0,$$

so M is positive definite. Similarly,

$$z^T A z = \left\| \nabla \sum_j z_j \phi_j \right\|_{L^2}^2 > 0$$

under homogeneous Dirichlet boundary conditions, so A is also positive definite.

Since M is invertible,

$$MC' + AC = F$$

can be rewritten as

$$C' = -M^{-1}AC + M^{-1}F.$$

Thus the Galerkin spatial discretization reduces the parabolic PDE to a finite-dimensional system of ODEs.

For the homogeneous problem,

$$MC' + AC = 0.$$

Multiplying by C^T gives

$$C^T MC' + C^T AC = 0.$$

Since M is symmetric,

$$C^T MC' = \frac{1}{2}\frac{d}{dt}\left(C^T MC\right).$$

Therefore,

$$\frac{1}{2}\frac{d}{dt}\|u_N\|_{L^2}^2 + \|\nabla u_N\|_{L^2}^2 = 0.$$

Hence,

$$\boxed{\frac{d}{dt}\|u_N\|_{L^2}^2 \leq 0}.$$

Thus the semidiscrete Galerkin method preserves the dissipative structure of the heat equation.

9.3 Fourier Spectral Galerkin Method

For a periodic problem on

$$\Omega = [0, 1],$$

choose the Fourier basis

$$\phi_j(x) = e^{2\pi i j x}, \qquad j = -N, \ldots, N.$$

The Fourier modes are orthogonal:

$$(\phi_j, \phi_k) = \delta_{jk}.$$

Therefore,

$$M_{jk} = \delta_{jk},$$

so

$$\boxed{M = I}.$$

Since

$$\phi_j'(x) = 2\pi i j \phi_j(x),$$

we have

$$A_{jk} = (\phi_j', \phi_k') = 4\pi^2 jk\, \delta_{jk}.$$

Since $\delta_{jk} = 0$ unless $j = k$,

$$A_{jk} = 4\pi^2 j^2 \delta_{jk}.$$

Thus A is diagonal.

Define

$$\widehat{f}_j(t) = \int_0^1 f(x, t) e^{-2\pi i j x}\, dx.$$

Then each Fourier coefficient satisfies

$$\boxed{c_j'(t) + 4\pi^2 j^2 c_j(t) = \widehat{f}_j(t)}.$$

The initial condition is

$$c_j(0) = \widehat{u}_{0,j}.$$

Therefore, the Fourier spectral method transforms the PDE into a set of independent scalar ODEs. This simplification follows from the orthogonality of the Fourier basis and the fact that Fourier modes diagonalize the Laplacian.

9.4 Piecewise Linear Finite Element Semidiscretization

Now let V_N be the piecewise linear finite element space on

$$0 = x_0 < x_1 < \cdots < x_N < x_{N+1} = 1.$$

Define

$$h_i = x_i - x_{i-1},$$

and let $\phi_i(x)$ be the standard hat function centered at x_i.

The Galerkin approximation is

$$u_h(x,t) = \sum_{i=1}^{N} c_i(t)\phi_i(x),$$

and the semidiscrete system again has the form

$$\boxed{MC'(t) + AC(t) = F(t)}.$$

The mass matrix is

$$M_{ij} = \int_0^1 \phi_i(x)\phi_j(x)\,dx.$$

Since two hat functions overlap only when

$$|i-j| \le 1,$$

we have

$$M_{ij} = \begin{cases} \dfrac{h_i + h_{i+1}}{3}, & i = j, \\ \dfrac{h_i}{6}, & i = j+1, \\ \dfrac{h_{i+1}}{6}, & i = j-1, \\ 0, & \text{otherwise.} \end{cases}$$

The stiffness matrix is

$$A_{ij} = \int_0^1 \phi_i'(x)\phi_j'(x)\,dx,$$

so

$$A_{ij} = \begin{cases} \dfrac{1}{h_i} + \dfrac{1}{h_{i+1}}, & i = j, \\ -\dfrac{1}{h_i}, & i = j+1, \\ -\dfrac{1}{h_{i+1}}, & i = j-1, \\ 0, & \text{otherwise.} \end{cases}$$

Thus both M and A are sparse and tridiagonal.

For a uniform mesh,

$$h_i = h,$$

we obtain

$$M = \frac{h}{6}\begin{bmatrix} 4 & 1 & & \\ 1 & 4 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & 4 \end{bmatrix},$$

and

$$A = \frac{1}{h}\begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{bmatrix}.$$

After spatial discretization, any ODE time-stepping method from Chapter 7 can be applied to

$$MC' + AC = F.$$

For Forward Euler,

$$M\frac{C^{n+1} - C^n}{\tau} + AC^n = F^n.$$

Therefore,

$$MC^{n+1} = (M - \tau A)C^n + \tau F^n.$$

For Backward Euler,

$$M\frac{C^{n+1} - C^n}{\tau} + AC^{n+1} = F^{n+1}.$$

Hence,

$$\boxed{(M + \tau A)C^{n+1} = MC^n + \tau F^{n+1}}.$$

For Crank–Nicolson,

$$M\frac{C^{n+1} - C^n}{\tau} + \frac{1}{2}A\left(C^{n+1} + C^n\right) = \frac{1}{2}\left(F^{n+1} + F^n\right).$$

Therefore,

$$\boxed{\left(M + \frac{\tau}{2}A\right)C^{n+1} = \left(M - \frac{\tau}{2}A\right)C^n + \frac{\tau}{2}\left(F^{n+1} + F^n\right)}.$$

9.5 Semilinear Parabolic Equations

Consider the semilinear equation

$$\boxed{u_t = \Delta u + f(u)}.$$

The nonlinear term creates an additional difficulty because a fully implicit discretization would require solving a nonlinear system at each time step.

The notes use a semi-implicit method: treat the diffusion term implicitly and the nonlinear term explicitly.

Define

$$\delta_x^2 U_j = \frac{U_{j-1} - 2U_j + U_{j+1}}{h^2}.$$

The numerical method is

$$\boxed{U_j^{n+1} = U_j^n + \tau\delta_x^2 U_j^{n+1} + \tau f(U_j^n)}.$$

Equivalently,

$$\left(I - \tau\delta_x^2\right)U^{n+1} = U^n + \tau f(U^n).$$

To estimate the error, substitute the exact solution:

$$u(x_j, t_{n+1}) = u(x_j, t_n) + \tau f(u(x_j, t_n)) + \tau\delta_x^2 u(x_j, t_{n+1}) + \xi_j^{n+1}.$$

For a sufficiently smooth solution,

$$\boxed{\xi_j^{n+1} = O\left(\tau(\tau + h^2)\right)}.$$

Define the global error

$$e_j^n = u(x_j, t_n) - U_j^n.$$

Subtracting the numerical equation from the exact discrete equation gives

$$e_j^{n+1} = e_j^n + \tau \delta_x^2 e_j^{n+1} + \tau \left[f(u(x_j, t_n)) - f(U_j^n) \right] + \xi_j^{n+1}.$$

Assume that f is Lipschitz continuous:

$$|f(u) - f(v)| \le L|u - v|.$$

By the mean value theorem,

$$f(u(x_j, t_n)) - f(U_j^n) = f'(\eta_j^n) e_j^n$$

for some intermediate value η_j^n. Therefore,

$$e^{n+1} = \left(I - \tau \delta_x^2 \right)^{-1} \left[e^n + \tau f'(\eta^n) e^n + \xi^{n+1} \right].$$

Using

$$\left\| \left(I - \tau \delta_x^2 \right)^{-1} \right\|_\infty \le 1,$$

we obtain

$$\|e^{n+1}\|_\infty \le (1 + L\tau) \|e^n\|_\infty + C\tau(\tau + h^2).$$

Iterating gives

$$\boxed{\|e^n\|_\infty \le C e^{L t_n} (\tau + h^2)}.$$

Thus the semi-implicit method is first-order accurate in time and second-order accurate in space. :contentReference[oaicite:2]index=2

9.6 Galerkin Approximation of the Nonlinear Problem

For

$$u_t = \Delta u + f(u),$$

the Galerkin approximation seeks

$$u_N(t) \in V_N$$

such that

$$(\partial_t u_N, v_N) + (\nabla u_N, \nabla v_N) = (f(u_N), v_N)$$

for every

$$v_N \in V_N.$$

Write

$$u_N(x, t) = \sum_{j=1}^{N} c_j(t) \phi_j(x).$$

In general,

$$f(u_N) \notin V_N.$$

Therefore, as emphasized in the notes, project $f(u_N)$ onto

$$\operatorname{span} \{ \phi_i \}.$$

Define

$$I_N f(u_N) = \sum_{j=1}^{N} f_j(t)\phi_j(x)$$

by

$$\boxed{(I_N f(u_N), \phi_i) = (f(u_N), \phi_i)}$$

for every i. Thus,

$$\sum_{j=1}^{N} f_j(t)(\phi_j, \phi_i) = (f(u_N), \phi_i).$$

The Galerkin equations become

$$\sum_{j=1}^{N} c_j'(t)(\phi_j, \phi_i) + \sum_{j=1}^{N} c_j(t)(\nabla\phi_j, \nabla\phi_i)$$

$$= \sum_{j=1}^{N} f_j(t)(\phi_j, \phi_i).$$

In matrix form,

$$\boxed{MC' + AC = Mf_N}.$$

Using the same semi-implicit idea as in the finite difference method,

$$M\frac{C^{n+1} - C^n}{\tau} + AC^{n+1} = Mf_N^n.$$

Therefore,

$$(M + \tau A)C^{n+1} = M\left(C^n + \tau f_N^n\right).$$

Hence,

$$\boxed{C^{n+1} = (M + \tau A)^{-1} M\left(C^n + \tau f_N^n\right)}.$$

Thus the same basic principle applies in both finite difference and Galerkin discretizations:

$$\begin{aligned} \text{diffusion term}: &\quad \text{implicit}, \\ \text{nonlinear term}: &\quad \text{explicit}. \end{aligned}$$

The resulting scheme avoids solving a nonlinear system at every time step while retaining the stability advantage of an implicit treatment of the diffusion operator.

Chapter 10

Numerical Methods for Hyperbolic Equations

10.1 Linear Advection and Characteristics

Consider the Cauchy problem

$$\begin{cases} u_t + a u_x = 0, & t > 0, \quad x \in (-\infty, +\infty), \\ u(x,0) = u_0(x). \end{cases}$$

The exact solution is

$$\boxed{u(x,t) = u_0(x - at)}.$$

Thus the initial profile is transported with constant velocity a without changing its shape.

The characteristic curves satisfy

$$\frac{dx}{dt} = a.$$

Hence,

$$x - at = C.$$

Along a characteristic,

$$\frac{d}{dt} u(x(t), t) = u_t + \frac{dx}{dt} u_x = u_t + a u_x = 0.$$

Therefore, u remains constant along each characteristic, which again gives

$$u(x,t) = u_0(x - at).$$

Consider now the bounded-domain problem

$$\begin{cases} u_t + a u_x = 0, & t \in (0, T], \quad x \in (0,1), \\ u(x,0) = u_0(x), & x \in (0,1), \\ u(0,t) = g(t). \end{cases}$$

Assume

$$a > 0.$$

Tracing the characteristic backward from (x,t) gives

$$u(x,t) = \begin{cases} u_0(x - at), & 0 \le x - at \le 1, \\ g\left(t - \dfrac{x}{a}\right), & x - at < 0. \end{cases}$$

Thus, for $a > 0$, information enters the domain through $x = 0$, so the boundary condition should be imposed there. For $a < 0$, the inflow boundary is $x = 1$.

Remark 10.1.1. For a hyperbolic equation, boundary conditions are determined by the direction of information propagation. Data should be prescribed at the inflow boundary rather than arbitrarily at both endpoints.

10.2 Centered Semidiscretization and Conservation

Consider the periodic problem

$$\begin{cases} u_t + au_x = 0, & t \in (0,T], \quad x \in (0,1), \\ u(x,0) = u_0(x), \\ u(0,t) = u(1,t). \end{cases}$$

Introduce the mesh

$$h = \frac{1}{M}, \qquad x_j = jh, \qquad j = 1, 2, \ldots, M.$$

Let

$$U_j(t) \approx u(x_j, t),$$

with periodic indexing.

Using the centered difference,

$$u_x(x_j,t) \approx \frac{U_{j+1}(t) - U_{j-1}(t)}{2h},$$

we obtain

$$\frac{dU_j}{dt} = -\frac{a}{2h}\left(U_{j+1} - U_{j-1}\right).$$

In matrix form,

$$\boxed{U' = AU},$$

where A is the periodic centered-difference matrix.

The matrix satisfies

$$A^T = -A.$$

Therefore,

$$A^T A = -A^2,$$

and all eigenvalues of A are purely imaginary.

The continuous equation conserves the L^2-norm. Multiply

$$u_t + au_x = 0$$

by u and integrate:

$$\int_0^1 uu_t\,dx + a\int_0^1 uu_x\,dx = 0.$$

Hence,

$$\frac{1}{2}\frac{d}{dt}\|u\|_{L^2}^2 + \frac{a}{2}u^2\Big|_0^1 = 0.$$

Under periodic boundary conditions,

$$u(0,t) = u(1,t),$$

so

$$\boxed{\|u(\cdot,t)\|_{L^2} = \|u(\cdot,0)\|_{L^2}}.$$

The semidiscrete equation has the same property. Since

$$U' = AU,$$

we have

$$\frac{d}{dt}\left(U^T U\right) = 2U^T AU.$$

Since $A^T = -A$,

$$U^T AU = (U^T AU)^T = U^T A^T U = -U^T AU.$$

Therefore,

$$U^T AU = 0.$$

Hence,

$$\boxed{\|U(t)\|_2 = \|U(0)\|_2}.$$

Thus the centered spatial semidiscretization preserves the discrete L^2-norm exactly.

10.3 Time Discretization and Stability

Introduce

$$\tau = \frac{T}{N}, \qquad t_n = n\tau,$$

and let

$$U_j^n \approx U_j(t_n) \approx u(x_j, t_n).$$

Forward Euler applied to

$$U' = AU$$

gives

$$U^{n+1} = U^n + \tau AU^n.$$

Therefore,

$$\boxed{U^{n+1} = (I + \tau A)U^n}.$$

At the grid level,

$$U_j^{n+1} = U_j^n - \frac{a\tau}{2h}\left(U_{j+1}^n - U_{j-1}^n\right).$$

Backward Euler gives

$$U^{n+1} - \tau AU^{n+1} = U^n.$$

Hence,

$$\boxed{U^{n+1} = (I - \tau A)^{-1} U^n}.$$

At the grid level,

$$U_j^{n+1} + \frac{a\tau}{2h}\left(U_{j+1}^{n+1} - U_{j-1}^{n+1}\right) = U_j^n.$$

Crank–Nicolson gives

$$U^{n+1} - \frac{\tau}{2}AU^{n+1} = U^n + \frac{\tau}{2}AU^n.$$

Therefore,

$$\boxed{U^{n+1} = \left(I - \frac{\tau}{2}A\right)^{-1}\left(I + \frac{\tau}{2}A\right)U^n}.$$

For Forward Euler with centered space difference, the local truncation error is

$$\xi_j^{n+1} = u(x_j, t_{n+1}) - u(x_j, t_n) + \frac{a\tau}{2h}\left[u(x_{j+1}, t_n) - u(x_{j-1}, t_n)\right].$$

Taylor expansion gives

$$u(x_j, t_{n+1}) = u + \tau u_t + \frac{\tau^2}{2} u_{tt} + O(\tau^3),$$

and

$$\frac{u(x_{j+1}, t_n) - u(x_{j-1}, t_n)}{2h} = u_x + \frac{h^2}{6} u_{xxx} + O(h^4).$$

Since

$$u_t + a u_x = 0,$$

we obtain

$$\xi_j^{n+1} = \tau \left[\frac{\tau}{2} u_{tt} + \frac{a h^2}{6} u_{xxx} + O(\tau^2) + O(h^4) \right].$$

Therefore,

$$\boxed{\xi_j^{n+1} = O\left(\tau(\tau + h^2)\right)}.$$

For the periodic centered-difference matrix, the eigenvectors are

$$v_j^{(p)} = e^{2\pi i p j h},$$

and the corresponding eigenvalues are

$$\boxed{\lambda_p = -\frac{ia}{h} \sin(2\pi p h)}.$$

Thus the eigenvalues are purely imaginary.

For Forward Euler,

$$B_{\mathrm{FE}} = I + \tau A.$$

Hence,

$$\lambda_p(B_{\mathrm{FE}}) = 1 + \tau \lambda_p(A).$$

Therefore,

$$|\lambda_p(B_{\mathrm{FE}})|^2 = 1 + \frac{a^2 \tau^2}{h^2} \sin^2(2\pi p h) > 1$$

for nonzero modes. Thus Forward Euler with centered differences is not strongly stable.

The notes observe that Lax–Richtmyer stability would require

$$\frac{a^2 \tau^2}{h^2} \lesssim \tau,$$

which leads to the impractical restriction

$$\tau \lesssim \frac{h^2}{a^2}.$$

For Backward Euler,

$$B_{\mathrm{BE}} = (I - \tau A)^{-1}.$$

For every Fourier mode,

$$|\lambda_p(B_{\mathrm{BE}})|^2 = \frac{1}{1 + \dfrac{a^2 \tau^2}{h^2} \sin^2(2\pi p h)} \leq 1.$$

Therefore,

$$\boxed{\|B_{\mathrm{BE}}\|_2 \le 1}$$

for every $\tau > 0$.

For Crank–Nicolson,

$$B_{\mathrm{CN}} = \left(I - \frac{\tau}{2}A\right)^{-1}\left(I + \frac{\tau}{2}A\right).$$

Since the eigenvalues of A are purely imaginary,

$$|\lambda_p(B_{\mathrm{CN}})| = 1.$$

Hence,

$$\boxed{\|B_{\mathrm{CN}}\|_2 = 1}.$$

Thus Backward Euler and Crank–Nicolson are unconditionally strongly stable, while centered Forward Euler is not strongly stable.

10.4 Numerical Viscosity and Stabilized Schemes

Since centered Forward Euler is unstable, one approach is to add artificial diffusion:

$$u_t^\varepsilon + a u_x^\varepsilon = \varepsilon u_{xx}^\varepsilon,$$

where

$$u^\varepsilon \longrightarrow u$$

as

$$\varepsilon \longrightarrow 0.$$

Using Forward Euler in time and centered differences in space gives

$$\frac{U_j^{n+1} - U_j^n}{\tau} = -\frac{a}{2h}\left(U_{j+1}^n - U_{j-1}^n\right) + \frac{\varepsilon}{h^2}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right).$$

Thus the second-difference term acts as numerical viscosity.

The Lax–Friedrichs method replaces U_j^n in centered Forward Euler by the average

$$\frac{1}{2}\left(U_{j-1}^n + U_{j+1}^n\right).$$

Therefore,

$$\boxed{U_j^{n+1} = \frac{1}{2}\left(U_{j-1}^n + U_{j+1}^n\right) - \frac{a\tau}{2h}\left(U_{j+1}^n - U_{j-1}^n\right)}.$$

Subtracting U_j^n and dividing by τ gives

$$\frac{U_j^{n+1} - U_j^n}{\tau} + a\frac{U_{j+1}^n - U_{j-1}^n}{2h} = \frac{h^2}{2\tau}\frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{h^2}.$$

Thus the Lax–Friedrichs numerical viscosity is

$$\boxed{\varepsilon_{\mathrm{LF}} = \frac{h^2}{2\tau}}.$$

The strong-stability condition is

$$\boxed{\left|\frac{a\tau}{h}\right| \le 1}.$$

The local truncation error recorded in the notes is

$$O\left(\tau(\tau + h^2)\right)$$

under the corresponding scaling.

The Lax–Wendroff method is obtained from a second-order Taylor expansion:

$$u(x_j, t_{n+1}) = u(x_j, t_n) + \tau u_t + \frac{\tau^2}{2} u_{tt} + O(\tau^3).$$

Since

$$u_t = -a u_x,$$

we have

$$u_{tt} = a^2 u_{xx}.$$

Therefore,

$$u(x_j, t_{n+1}) = u - a\tau u_x + \frac{a^2\tau^2}{2} u_{xx} + O(\tau^3).$$

Approximating the spatial derivatives gives

$$\boxed{U_j^{n+1} = U_j^n - \frac{a\tau}{2h}\left(U_{j+1}^n - U_{j-1}^n\right) + \frac{a^2\tau^2}{2h^2}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right)}.$$

Thus the effective viscosity coefficient is

$$\boxed{\varepsilon_{\text{LW}} = \frac{a^2\tau}{2}}.$$

The strong-stability condition is again

$$\boxed{\left|\frac{a\tau}{h}\right| \le 1}.$$

The local truncation error is

$$O\left(\tau(\tau^2 + h^2)\right).$$

Remark 10.4.1. Both Lax–Friedrichs and Lax–Wendroff can be viewed as centered Forward Euler schemes with additional terms. In Lax–Friedrichs the added term is strongly diffusive, while in Lax–Wendroff it arises from the second-order Taylor correction.

10.5 Upwind and Beam–Warming Methods

Centered differences ignore the direction of information propagation. Upwind methods use the characteristic direction explicitly.

The one-sided first differences are

$$u_x(x_j) \approx \frac{u(x_j) - u(x_{j-1})}{h}$$

from the left and

$$u_x(x_j) \approx \frac{u(x_{j+1}) - u(x_j)}{h}$$

from the right.

If

$$a > 0,$$

then

$$u(x_j, t + \tau) = u(x_j - a\tau, t),$$

so the relevant information lies to the left of x_j. Therefore,

$$\boxed{U_j^{n+1} = U_j^n - \frac{a\tau}{h}\left(U_j^n - U_{j-1}^n\right)}.$$

If

$$a < 0,$$

the relevant information lies to the right, so

$$\boxed{U_j^{n+1} = U_j^n - \frac{a\tau}{h}\left(U_{j+1}^n - U_j^n\right)}.$$

For $a > 0$, the method can be rewritten as

$$U_j^{n+1} = U_j^n - \frac{a\tau}{2h}\left(U_{j+1}^n - U_{j-1}^n\right) + \frac{a\tau}{2h}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right).$$

Thus the upwind method contains numerical viscosity

$$\boxed{\varepsilon_{\text{UW}} = \frac{|a|h}{2}}.$$

The strong-stability condition is

$$\boxed{\left|\frac{a\tau}{h}\right| \le 1}.$$

Its local truncation error is

$$\boxed{O\left(\tau(\tau + h)\right)}.$$

A second-order upwind-biased scheme is the Beam–Warming method. For $a > 0$,

$$\boxed{U_j^{n+1} = U_j^n - \frac{a\tau}{2h}\left(3U_j^n - 4U_{j-1}^n + U_{j-2}^n\right) + \frac{a^2\tau^2}{2h^2}\left(U_j^n - 2U_{j-1}^n + U_{j-2}^n\right)}.$$

Define

$$\mu = \frac{a\tau}{h}.$$

Then

$$U_j^{n+1} = U_j^n - \frac{\mu}{2}\left(3U_j^n - 4U_{j-1}^n + U_{j-2}^n\right) + \frac{\mu^2}{2}\left(U_j^n - 2U_{j-1}^n + U_{j-2}^n\right).$$

Equivalently,

$$U_j^{n+1} = \left(1 - \frac{3}{2}\mu + \frac{1}{2}\mu^2\right)U_j^n + \left(2\mu - \mu^2\right)U_{j-1}^n + \frac{1}{2}\left(\mu^2 - \mu\right)U_{j-2}^n.$$

The strong-stability condition recorded in the notes is

$$\boxed{0 \le \mu \le 2}.$$

The local truncation error is

$$O\left(\tau(\tau^2 + \tau h + h^2)\right).$$

If

$$\tau \approx h,$$

the method is second-order accurate.

10.6 Von Neumann and Interpolation Analysis

Von Neumann analysis gives a unified derivation of the strong-stability conditions. Assume

$$U_j^n = g(\xi)^n e^{ij\xi h}.$$

Strong stability requires

$$\boxed{|g(\xi)| \le 1}$$

for every frequency ξ.

For centered Forward Euler,

$$g_{\mathrm{FE}}(\xi) = 1 - i\frac{a\tau}{h}\sin(\xi h).$$

Therefore,

$$|g_{\mathrm{FE}}(\xi)|^2 = 1 + \frac{a^2\tau^2}{h^2}\sin^2(\xi h) \ge 1.$$

Hence the scheme is not strongly stable.

For Lax–Friedrichs,

$$g_{\mathrm{LF}}(\xi) = \cos(\xi h) - i\frac{a\tau}{h}\sin(\xi h).$$

Thus,

$$|g_{\mathrm{LF}}(\xi)|^2 = \cos^2(\xi h) + \frac{a^2\tau^2}{h^2}\sin^2(\xi h).$$

Therefore,

$$|g_{\mathrm{LF}}(\xi)| \le 1$$

provided

$$\boxed{\left|\frac{a\tau}{h}\right| \le 1}.$$

For Lax–Wendroff,

$$g_{\mathrm{LW}}(\xi) = 1 - i\frac{a\tau}{h}\sin(\xi h) + \frac{a^2\tau^2}{h^2}\left(\cos(\xi h) - 1\right).$$

A direct calculation gives

$$|g_{\mathrm{LW}}(\xi)|^2 = 1 - 4\mu^2(1-\mu^2)\sin^4\left(\frac{\xi h}{2}\right),$$

where

$$\mu = \frac{a\tau}{h}.$$

Hence,

$$|g_{\mathrm{LW}}(\xi)| \le 1$$

when

$$\boxed{|\mu| \le 1}.$$

For the upwind method with $a > 0$,

$$g_{\mathrm{UW}}(\xi) = 1 - \mu\left(1 - e^{-i\xi h}\right).$$

Therefore,

$$|g_{\mathrm{UW}}(\xi)|^2 = 1 - 4\mu(1-\mu)\sin^2\left(\frac{\xi h}{2}\right).$$

Hence,

$$|g_{\mathrm{UW}}(\xi)| \le 1$$

when

$$\boxed{0 \le \mu \le 1}.$$

The same schemes can also be interpreted through interpolation. Since

$$u(x_j, t_{n+1}) = u(x_j - a\tau, t_n),$$

we approximate the value at $x_j - a\tau$ using nearby grid values.

For $a > 0$, linear interpolation on

$$[x_{j-1}, x_j]$$

gives

$$P_1(x) = U_{j-1}^n \frac{x_j - x}{h} + U_j^n \frac{x - x_{j-1}}{h}.$$

Evaluating at

$$x = x_j - a\tau$$

gives

$$U_j^{n+1} = P_1(x_j - a\tau) = U_j^n - \frac{a\tau}{h}\left(U_j^n - U_{j-1}^n\right),$$

which is exactly the upwind method.

Quadratic interpolation using

$$x_{j-1}, x_j, x_{j+1}$$

gives the Lax–Wendroff method:

$$U_j^{n+1} = U_j^n - \frac{a\tau}{2h}\left(U_{j+1}^n - U_{j-1}^n\right) + \frac{a^2\tau^2}{2h^2}\left(U_{j-1}^n - 2U_j^n + U_{j+1}^n\right).$$

Quadratic interpolation using the upwind points

$$x_{j-2}, x_{j-1}, x_j$$

gives the Beam–Warming method:

$$U_j^{n+1} = U_j^n - \frac{a\tau}{2h}\left(3U_j^n - 4U_{j-1}^n + U_{j-2}^n\right) + \frac{a^2\tau^2}{2h^2}\left(U_j^n - 2U_{j-1}^n + U_{j-2}^n\right).$$

The notes emphasize the principle

$$\boxed{\text{Interpolation is stable, while extrapolation is unstable.}}$$

For example, if

$$\frac{a\tau}{h} > 1,$$

then $x_j - a\tau$ lies outside $[x_{j-1}, x_j]$, so the upwind formula becomes an extrapolation rather than an interpolation.

10.7 Domain of Dependence and the CFL Condition

Definition 10.7.1 (Domain of Dependence)**.** The domain of dependence $D(x,T)$ is the set of points in the initial data on which the exact value $u(x,T)$ depends.

For

$$u_t + au_x = 0,$$

we have

$$u(x,T) = u_0(x - aT),$$

so

$$\boxed{D(x,T) = \{x - aT\}}.$$

For the hyperbolic system

$$U_t + AU_x = 0,$$

if λ_p are the eigenvalues of A, then

$$D(x,T) = \{x - \lambda_p T : p = 1, 2, \ldots, s\}.$$

For the heat equation,

$$u_t = u_{xx},$$

the domain of dependence is

$$\boxed{D(x,T) = (-\infty, +\infty)}.$$

The numerical domain of dependence $D^N(x,T)$ consists of the initial grid points that influence the numerical value at (x,T).

For example, in a nearest-neighbor method such as Lax–Wendroff, after n time steps,

$$U_j^n$$

depends on

$$U_{j-n}^0, U_{j-n+1}^0, \ldots, U_j^0, \ldots, U_{j+n}^0.$$

Let

$$x = x_j, \qquad T = t_n = n\tau.$$

Then

$$x_{j\pm n} = x_j \pm nh = x \pm \frac{T}{\tau}h.$$

Therefore, in the limit,

$$D^N(x,T) = \left[x - \frac{Th}{\tau}, x + \frac{Th}{\tau}\right].$$

For the exact characteristic point $x - aT$ to lie inside the numerical domain of dependence, we need

$$|a|T \leq \frac{Th}{\tau}.$$

Hence,

$$\boxed{\left|\frac{a\tau}{h}\right| \leq 1}.$$

Theorem 10.7.2 (CFL Condition)**.** *A numerical method can be convergent only if the numerical domain of dependence contains the exact domain of dependence, at least in the limit as*

$$\tau, h \longrightarrow 0.$$

Symbolically,

$$\boxed{D(x,T) \subseteq D^N(x,T)}.$$

For Lax–Friedrichs and Lax–Wendroff, the CFL condition is

$$\left|\frac{a\tau}{h}\right| \leq 1.$$

For the upwind method with $a > 0$, the numerical domain extends only to the left, and the condition is

$$0 \leq \frac{a\tau}{h} \leq 1.$$

For Beam–Warming with $a > 0$, the two-cell upwind stencil gives

$$\boxed{0 \leq \frac{a\tau}{h} \leq 2}.$$

Remark 10.7.3. The CFL condition is a necessary condition for convergence, not a sufficient condition. A method may have a sufficiently large numerical domain of dependence and still fail because it is unstable or inconsistent.

For the heat equation, the exact domain of dependence is the entire real line. If Forward Euler is used with a fixed ratio

$$\frac{\tau}{h} = \gamma,$$

then the numerical domain remains bounded as $h, \tau \to 0$, so the CFL requirement cannot be satisfied.

If instead

$$\frac{\tau}{h^2} = \gamma,$$

then

$$\frac{h}{\tau} = \frac{1}{\gamma h} \longrightarrow +\infty,$$

so the numerical domain expands to the whole real line. This is consistent with the familiar parabolic scaling

$$\tau = O(h^2).$$

Backward Euler has an infinite numerical domain of dependence at each time step because solving the implicit linear system couples all grid points. Therefore, it satisfies the domain-of-dependence requirement for every

$$\tau > 0.$$

Chapter 11

Conservation Laws and Structure-Preserving Methods

11.1 Nonlinear Conservation Laws

Consider the scalar conservation law

$$u_t + (f(u))_x = 0.$$

Define

$$a(u) = f'(u).$$

For a smooth solution,

$$(f(u))_x = f'(u)u_x = a(u)u_x,$$

so the equation may also be written as

$$u_t + a(u)u_x = 0.$$

Integrating over an interval $I = [a, b]$ gives

$$\int_a^b u_t \, dx + \int_a^b (f(u))_x \, dx = 0.$$

Therefore,

$$\frac{d}{dt} \int_a^b u(x,t) \, dx = f(u(a,t)) - f(u(b,t)).$$

Thus the total amount of u changes only through fluxes across the boundary.

This conservation property motivates numerical schemes of the form

$$\boxed{U_j^{n+1} = U_j^n - \frac{\tau}{h}\left(F_{j+\frac{1}{2}}^n - F_{j-\frac{1}{2}}^n\right)}.$$

When the equations are summed over consecutive grid cells, all interior numerical fluxes cancel.

This flux-difference form is particularly important for nonlinear conservation laws.

11.2 Conservative Numerical Schemes

The Lax–Friedrichs method for

$$u_t + (f(u))_x = 0$$

is

$$\boxed{U_j^{n+1} = \frac{1}{2}\left(U_{j-1}^n + U_{j+1}^n\right) - \frac{\tau}{2h}\left[f(U_{j+1}^n) - f(U_{j-1}^n)\right]}.$$

Define

$$g_1(x) = \frac{1}{2}\left(x - \frac{\tau}{h} f(x)\right)$$

and

$$g_2(x) = \frac{1}{2}\left(x + \frac{\tau}{h} f(x)\right).$$

Then

$$U_j^{n+1} = g_1(U_{j+1}^n) + g_2(U_{j-1}^n).$$

We have

$$g_1'(x) = \frac{1}{2}\left(1 - \frac{\tau}{h} f'(x)\right),$$

and

$$g_2'(x) = \frac{1}{2}\left(1 + \frac{\tau}{h} f'(x)\right).$$

Therefore, both functions are monotone if

$$\left|\frac{\tau}{h} f'(x)\right| \le 1.$$

Since

$$a(u) = f'(u),$$

a sufficient condition is

$$\boxed{\frac{\tau}{h} \max_j |a(U_j^n)| \le 1}.$$

Now consider an upwind discretization. Suppose

$$a(u) > 0.$$

A direct discretization of

$$u_t + a(u)u_x = 0$$

is

$$U_j^{n+1} = U_j^n - \frac{\tau}{h} a(U_j^n)\left(U_j^n - U_{j-1}^n\right).$$

However, this is not generally conservative because

$$a(U_j)\left(U_j - U_{j-1}\right)$$

is not exactly a difference of numerical fluxes.

The conservative version is

$$\boxed{U_j^{n+1} = U_j^n - \frac{\tau}{h}\left[f(U_j^n) - f(U_{j-1}^n)\right]}.$$

Remark 11.2.1. For a linear flux

$$f(u) = au,$$

the conservative and nonconservative upwind forms coincide. For a nonlinear flux, this is no longer true.

11.3 Lax–Wendroff Method for Nonlinear Conservation Laws

For

$$u_t + (f(u))_x = 0,$$

we have

$$u_t = -(f(u))_x.$$

Differentiate with respect to time:

$$u_{tt} = -(f(u)_t)_x.$$

Since

$$f(u)_t = f'(u)u_t = a(u)u_t,$$

we obtain

$$u_{tt} = -\left(a(u)u_t\right)_x .$$

Using

$$u_t = -(f(u))_x,$$

gives

$$\boxed{u_{tt} = \left[a(u)(f(u))_x\right]_x}.$$

Taylor expansion in time gives

$$u(x_j, t_{n+1}) = u(x_j, t_n) + \tau u_t(x_j, t_n) + \frac{\tau^2}{2} u_{tt}(x_j, t_n) + O(\tau^3).$$

Therefore,

$$u(x_j, t_{n+1}) = u - \tau(f(u))_x + \frac{\tau^2}{2}\left[a(u)(f(u))_x\right]_x + O(\tau^3).$$

Let

$$f_j^n = f(U_j^n).$$

Approximate the first derivative by

$$(f(u))_x \approx \frac{f_{j+1}^n - f_{j-1}^n}{2h}.$$

For the second term, define

$$a_{j+\frac{1}{2}}^n = a(U_{j+\frac{1}{2}}^n).$$

Then

$$\left[a(u)(f(u))_x\right]_x$$

is approximated by

$$\frac{1}{h^2}\left[a_{j+\frac{1}{2}}^n \left(f_{j+1}^n - f_j^n\right) - a_{j-\frac{1}{2}}^n \left(f_j^n - f_{j-1}^n\right)\right].$$

Hence the nonlinear Lax–Wendroff method is

$$\boxed{\begin{aligned} U_j^{n+1} = U_j^n &- \frac{\tau}{2h}\left(f_{j+1}^n - f_{j-1}^n\right) \\ &+ \frac{\tau^2}{2h^2}\left[a_{j+\frac{1}{2}}^n \left(f_{j+1}^n - f_j^n\right) - a_{j-\frac{1}{2}}^n \left(f_j^n - f_{j-1}^n\right)\right]. \end{aligned}}$$

The notes use

$$a_{j+\frac{1}{2}}^n = a(U_{j+\frac{1}{2}}^n)$$

with the practical approximation

$$\boxed{a_{j+\frac{1}{2}}^n \approx a\left(\frac{U_j^n + U_{j+1}^n}{2}\right)}.$$

Thus the same Taylor-expansion idea used for the linear Lax–Wendroff method extends naturally to nonlinear conservation laws.

11.4 Variational Derivatives and Euler–Lagrange Equations

Consider the minimization problem

$$\min_{u \in V} E(u),$$

where

$$E(u) = \int_0^1 F(u, u')\, dx$$

and

$$V = \left\{ u \in H^1(0,1) : u|_{\partial\Omega} = g \right\}.$$

Let

$$V_0 = \left\{ \phi \in H^1(0,1) : \phi|_{\partial\Omega} = 0 \right\}.$$

If u is a minimizer, then

$$E(u + \varepsilon\phi) \geq E(u)$$

for every sufficiently small ε and every $\phi \in V_0$. Therefore,

$$\left.\frac{d}{d\varepsilon} E(u + \varepsilon\phi)\right|_{\varepsilon=0} = 0.$$

We have

$$E(u + \varepsilon\phi) = \int_0^1 F(u + \varepsilon\phi, u' + \varepsilon\phi')\, dx.$$

Differentiating gives

$$\int_0^1 (F_u \phi + F_{u'} \phi')\, dx = 0.$$

Integrating the second term by parts gives

$$\int_0^1 \left[F_u - \frac{d}{dx} F_{u'} \right] \phi \, dx = 0.$$

Therefore,

$$\boxed{F_u - \frac{d}{dx} F_{u'} = 0}.$$

This is the Euler–Lagrange equation.

The variational derivative is denoted by

$$\boxed{\frac{\delta E}{\delta u} = F_u - \frac{d}{dx} F_{u'}}.$$

It is characterized by

$$\left.\frac{d}{d\varepsilon} E(u + \varepsilon\phi)\right|_{\varepsilon=0} = \left(\frac{\delta E}{\delta u}, \phi \right).$$

As a simple example, consider the shortest-distance functional

$$L(y) = \int_0^a \sqrt{1 + (y')^2}\, dx$$

with fixed endpoint values. Since

$$F(y, y') = \sqrt{1 + (y')^2},$$

the Euler–Lagrange equation gives

$$-\frac{d}{dx}\left(\frac{y'}{\sqrt{1 + (y')^2}}\right) = 0.$$

Therefore,

$$\frac{y'}{\sqrt{1 + (y')^2}} = C,$$

so y' is constant. Hence the minimizing curve is a straight line.

Another classical variational problem is the brachistochrone problem, which seeks the curve of fastest descent between two prescribed points. Its travel-time functional can also be treated by the Euler–Lagrange equation.

11.5 Allen–Cahn Equation and Structural Properties

Consider the Allen–Cahn equation

$$\boxed{u_t = \varepsilon \Delta u + \frac{1}{\varepsilon}\left(u - u^3\right)}.$$

Define the Ginzburg–Landau energy

$$\boxed{E(u) = \int_\Omega \left[\frac{\varepsilon}{2}|\nabla u|^2 + \frac{1}{\varepsilon}F(u)\right] dx},$$

where

$$F(u) = \frac{1}{4}\left(u^2 - 1\right)^2.$$

Since

$$F'(u) = u^3 - u,$$

the variational derivative is

$$\frac{\delta E}{\delta u} = -\varepsilon \Delta u + \frac{1}{\varepsilon}F'(u).$$

Therefore,

$$\boxed{u_t = -\frac{\delta E}{\delta u}}.$$

Thus the Allen–Cahn equation is the L^2-gradient flow of the Ginzburg–Landau energy.

Differentiate the energy along the solution:

$$\frac{d}{dt}E(u(t)) = \left(\frac{\delta E}{\delta u}, u_t\right).$$

Since

$$u_t = -\frac{\delta E}{\delta u},$$

we obtain

$$\boxed{\frac{d}{dt}E(u(t)) = -\|u_t\|_{L^2}^2 \leq 0}.$$

Therefore, the energy decreases as time increases.

The equation also satisfies a maximum bound principle.

Theorem 11.5.1 (Maximum Bound Principle)**.** *Assume*

$$|u(x,0)| \leq 1$$

and homogeneous Neumann boundary conditions

$$\frac{\partial u}{\partial n} = 0.$$

Then

$$\boxed{|u(x,t)| \leq 1}$$

for all $t > 0$.

The basic idea is to consider a positive interior maximum

$$u(x_m, t_m) = \max_x u(x, t_m).$$

At such a point,

$$\Delta u(x_m, t_m) \leq 0.$$

If

$$u(x_m, t_m) > 1,$$

then

$$u - u^3 < 0.$$

Hence,

$$u_t(x_m, t_m) = \varepsilon \Delta u + \frac{1}{\varepsilon}(u - u^3) < 0,$$

which prevents the maximum from crossing above 1. The lower bound is obtained analogously.

11.6 Maximum-Bound-Preserving Time Discretizations

We now ask whether the numerical method preserves

$$|U_j^n| \leq 1.$$

For Forward Euler,

$$U_j^{n+1} = U_j^n + \frac{\tau}{\varepsilon}\left[U_j^n - (U_j^n)^3\right] + \tau\varepsilon \frac{U_{j-1}^n - 2U_j^n + U_{j+1}^n}{h^2}.$$

Write

$$U_j^{n+1} = g(U_{j-1}^n, U_j^n, U_{j+1}^n).$$

Then

$$\frac{\partial g}{\partial U_{j-1}^n} = \frac{\tau\varepsilon}{h^2} > 0$$

and

$$\frac{\partial g}{\partial U_{j+1}^n} = \frac{\tau\varepsilon}{h^2} > 0.$$

Moreover,

$$\frac{\partial g}{\partial U_j^n} = 1 - \frac{2\tau\varepsilon}{h^2} + \frac{\tau}{\varepsilon}\left[1 - 3(U_j^n)^2\right].$$

For

$$|U_j^n| \le 1,$$

a sufficient monotonicity condition is

$$1 - \frac{2\tau\varepsilon}{h^2} - \frac{2\tau}{\varepsilon} \ge 0.$$

Thus Forward Euler requires a time-step restriction to preserve the maximum bound.

A semi-implicit method treats the diffusion term implicitly:

$$\boxed{U_j^{n+1} = U_j^n + \frac{\tau}{\varepsilon}\left[U_j^n - (U_j^n)^3\right] + \tau\varepsilon\delta_h^2 U_j^{n+1}}.$$

In vector form,

$$U^{n+1} = (I - \tau\varepsilon D_h)^{-1}\left[U^n + \frac{\tau}{\varepsilon}\left(U^n - (U^n)^3\right)\right].$$

Define

$$H(x) = x + \frac{\tau}{\varepsilon}(x - x^3).$$

Then

$$H(-1) = -1, \qquad H(1) = 1.$$

If H is increasing on $[-1, 1]$, then

$$|H(x)| \le 1$$

for every $x \in [-1, 1]$. Since

$$H'(x) = 1 + \frac{\tau}{\varepsilon}(1 - 3x^2),$$

we have

$$H'(x) \ge 0$$

on $[-1, 1]$ provided

$$\boxed{\tau \le \frac{\varepsilon}{2}}.$$

Thus the semi-implicit method preserves the maximum bound under this restriction.

Backward Euler would treat the nonlinear term implicitly as well, which requires solving a nonlinear system at each step.

The notes therefore introduce a stabilized semi-implicit method:

$$\boxed{\begin{aligned} U_j^{n+1} = U_j^n + \frac{\tau}{\varepsilon}\left[U_j^n - (U_j^n)^3\right] + \tau\varepsilon\delta_h^2 U_j^{n+1} \\ - \frac{\beta\tau}{\varepsilon}\left(U_j^{n+1} - U_j^n\right). \end{aligned}}$$

Rearranging gives

$$\left(1+\frac{\beta\tau}{\varepsilon}\right)U_j^{n+1}-\tau\varepsilon\delta_h^2U_j^{n+1}$$
$$=\left(1+\frac{\beta\tau}{\varepsilon}\right)U_j^n+\frac{\tau}{\varepsilon}\left[U_j^n-(U_j^n)^3\right].$$

Define

$$H_\beta(x)=\left(1+\frac{\beta\tau}{\varepsilon}\right)x+\frac{\tau}{\varepsilon}(x-x^3).$$

Then

$$H'_\beta(x)=1+\frac{\beta\tau}{\varepsilon}+\frac{\tau}{\varepsilon}(1-3x^2).$$

For

$$x\in[-1,1],$$

we have

$$H'_\beta(x)\geq 1+\frac{(\beta-2)\tau}{\varepsilon}.$$

Therefore, if

$$\boxed{\beta\geq 2},$$

then H_β is increasing on $[-1,1]$ for every $\tau>0$.

Moreover,

$$H_\beta(1)=1+\frac{\beta\tau}{\varepsilon},$$

and

$$H_\beta(-1)=-\left(1+\frac{\beta\tau}{\varepsilon}\right).$$

Using the discrete maximum principle for the implicit diffusion operator gives

$$\boxed{\|U^{n+1}\|_\infty\leq 1}$$

whenever

$$\|U^n\|_\infty\leq 1.$$

Thus, for

$$\beta\geq 2,$$

the stabilized semi-implicit method is unconditionally maximum-bound-preserving. :contentReference[oaicite:3]index=3

11.7 Discrete Energy Stability

Let A denote the discrete Laplacian matrix. Under the sign convention used here,

$$U^TAU\leq 0$$

for every grid vector U.

Define the discrete energy by

$$\boxed{E_h(U)=-\frac{\varepsilon}{2}U^TAU+\frac{h}{\varepsilon}\sum_j F(U_j)}.$$

Consider the stabilized semi-implicit scheme

$$\frac{U^{n+1}-U^n}{\tau}=\varepsilon AU^{n+1}-\frac{1}{\varepsilon}F'(U^n)-\frac{\beta}{\varepsilon}\left(U^{n+1}-U^n\right).$$

Take the discrete inner product with

$$U^{n+1}-U^n.$$

Then

$$\frac{1}{\tau}\left\|U^{n+1}-U^n\right\|^2=\varepsilon\left(AU^{n+1},U^{n+1}-U^n\right)$$
$$-\frac{1}{\varepsilon}\left(F'(U^n),U^{n+1}-U^n\right)-\frac{\beta}{\varepsilon}\left\|U^{n+1}-U^n\right\|^2.$$

Use the identity

$$(a,b-a)=\frac{1}{2}\left[(b,b)-(a,a)-(b-a,b-a)\right].$$

Applying this identity to the quadratic energy term and using Taylor expansion,

$$F(U_j^{n+1})=F(U_j^n)+F'(U_j^n)\left(U_j^{n+1}-U_j^n\right)$$
$$+\frac{1}{2}F''(\xi_j)\left(U_j^{n+1}-U_j^n\right)^2,$$

where ξ_j lies between U_j^n and U_j^{n+1}.

If the maximum bound

$$|U_j^n|\le 1,\qquad |U_j^{n+1}|\le 1$$

holds, then

$$|\xi_j|\le 1.$$

Since

$$F''(\xi)=3\xi^2-1,$$

we have

$$F''(\xi)\le 2$$

for

$$|\xi|\le 1.$$

Combining the quadratic and nonlinear terms gives

$$E_h(U^{n+1})-E_h(U^n)\le-\frac{\varepsilon}{2}\left(U^{n+1}-U^n\right)^T(-A)\left(U^{n+1}-U^n\right)$$
$$-\left(\frac{1}{\tau}+\frac{\beta-2}{2\varepsilon}\right)\left\|U^{n+1}-U^n\right\|^2.$$

Therefore, if

$$\boxed{\beta\ge 2},$$

then for every

$$\tau>0,$$

we obtain

$$\boxed{E_h(U^{n+1}) \leq E_h(U^n)}.$$

Thus the stabilization parameter $\beta \geq 2$ gives two important properties simultaneously:

$$\boxed{\|U^n\|_\infty \leq 1}$$

and

$$\boxed{E_h(U^{n+1}) \leq E_h(U^n)}.$$

Remark 11.7.1. The Allen–Cahn example illustrates a broader principle in numerical PDEs: accuracy and conventional stability are not always the only properties that matter. A useful discretization may also need to preserve structural properties of the continuous equation, such as a maximum principle, conservation law, positivity, or energy dissipation.

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Any errors, omissions, or inaccuracies in these notes are my own.